

M O D U L A R S Y S T E M

VECTORS IN THE PLANE

Ertuğrul Tarhan



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PREFACE

To the Teacher

When I taught mathematics in different schools, my students asked me why they were learning these topics and how they used them. These questions were important to me. While I was planning this book, I decided to answer my students' questions by including a good number of applications of vectors. I have written this book to teach vectors in the plane to high school students. The aim of the book is to introduce vectors in the plane in a way that students can understand.

This book has only one chapter, which is divided into three sections. In each section there are examples, self-check questions, and exercises.

The self-check questions (**Check Yourself** sections) check students' understanding at key points in the book. They can also be used in pop quizzes to check students' understanding of a subtopic such as the use of the polygon method for the addition of vectors. The aim of including applications is to show the use of vectors in different areas such as physics. **Examples** throughout the text help students to understand the topics and show how to use vectors to solve typical problems. **Exercises** cover the material in a whole section. They provide a thorough review of the material covered and are a means of checking students' general understanding of the topic. The summary at the end of the book lists the important ideas and concepts covered in the book. Finally, the multiple-choice tests show examples of multiple-choice exam questions and aim to prepare the students for such exams.

In the first section of the book, we analyze vectors and operations on vectors in synthetic geometry. In the second section, we analyze vectors and operations on vectors in analytic geometry. In the last section, we introduce the dot product and look at its use in different areas. As a result, I can say that if a student who is not familiar with vectors learns half of the material in this book, then he or she will have a good basic understanding of vectors. If the student learns all of the material in this book, then he or she will be able to study vectors in space and vectors in calculus easily at university.

The prerequisite for the material in this book is a basic familiarity with trigonometric ratios in right triangles, the cosine law, linear equations, real numbers, triangles, quadrilaterals, and polygons.

Acknowledgements

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Finally, I am grateful to my family for their support throughout the writing of this book. The success of the book will owe a great deal to their understanding and kindness.

Ertuğrul Tarhan

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INTRODUCTION

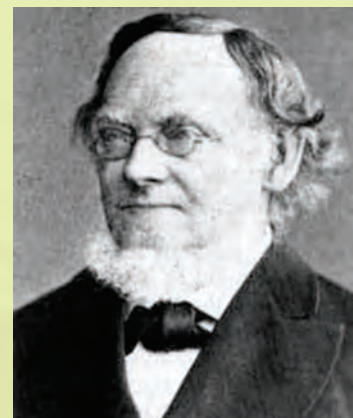


The origin of the parallelogram law for the addition of vectors is unknown. It may have appeared in a lost work of Aristotle (384-322 B.C.). It was also the first corollary in Isaac Newton's *Principia Mathematica* (1687). In the *Principia*, Newton (1642-1727) dealt extensively with vector quantities (e.g. velocity and force), but never the concept of a vector.

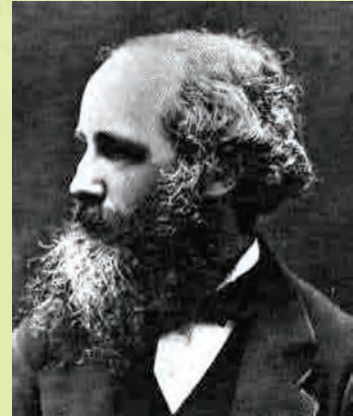
Vectors were born in the first two decades of the 19th century with the geometric representation of complex numbers. Caspar Wessel (1745-1818), Jean Robert Argand (1768-1822), Carl Friedrich Gauss (1777-1855), and others conceived of complex numbers as points in the two-dimensional plane, i.e. as two-dimensional vectors. Gauss used complex numbers to prove the Fundamental Theorem of Algebra (1799). In 1837, William Rowan Hamilton (1805-1865) showed that complex numbers could be considered abstractly as ordered pairs (a, b) of real numbers. This idea inspired many mathematicians, including Hamilton himself, to search for a way to extend the two-dimensional 'numbers' to three dimensions.

Hamilton gave up the search for such a three-dimensional system and instead invented a four-dimensional system that he called *quaternions*. He wrote two exhaustive books, *Lectures on Quaternions* (1853) and *Elements of Quaternions* (1866), which described the algebra of quaternions and how they could be used in geometry. Peter Guthrie Tait (1831-1901) began applying quaternions to problems in electricity and magnetism and to other problems in physics. In the second half of the 19th century, Tait's support of quaternions produced strong reactions in the scientific community.

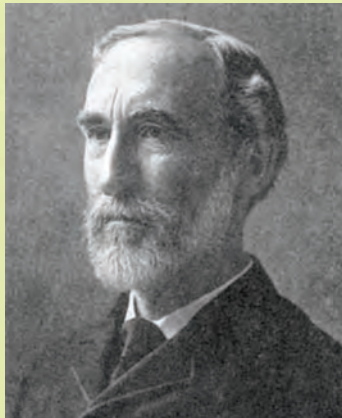
At about the same time that Hamilton discovered quaternions, Hermann Grassmann (1809-1877) was writing *The Calculus of Extension* (1844), now well known by its German title, *Ausdehnungslehre*. Unfortunately, *The Calculus of Extension* had two strikes against it. First, it was highly abstract, lacking in explanatory examples and written in a non-standard style with a very complicated notation. Second, Grassmann was a secondary school teacher without a major scientific reputation (compared to Hamilton). In 1862, Grassmann published a second and revised edition of his *The Calculus of Extension*, but it too was abstract for the mathematicians of the time, and it met essentially the same fate as his first edition. Finally, in the late 1860s and 1870s, *The Calculus of Extension* slowly began to be understood and appreciated. A third edition of *The Calculus of Extension* was published in 1878, the year after Grassmann died.



In his *Treatise on Electricity and Magnetism* (1873), Maxwell emphasized the importance of what he described as ‘quaternion ideas ... or the doctrine of Vectors’ as a ‘mathematical method.’ At the same time, he pointed out the inhomogeneous nature of the product of quaternions, and he warned scientists against using quaternion methods with its details involving the three vector components. Essentially, Maxwell was suggesting a purely vectorial analysis.

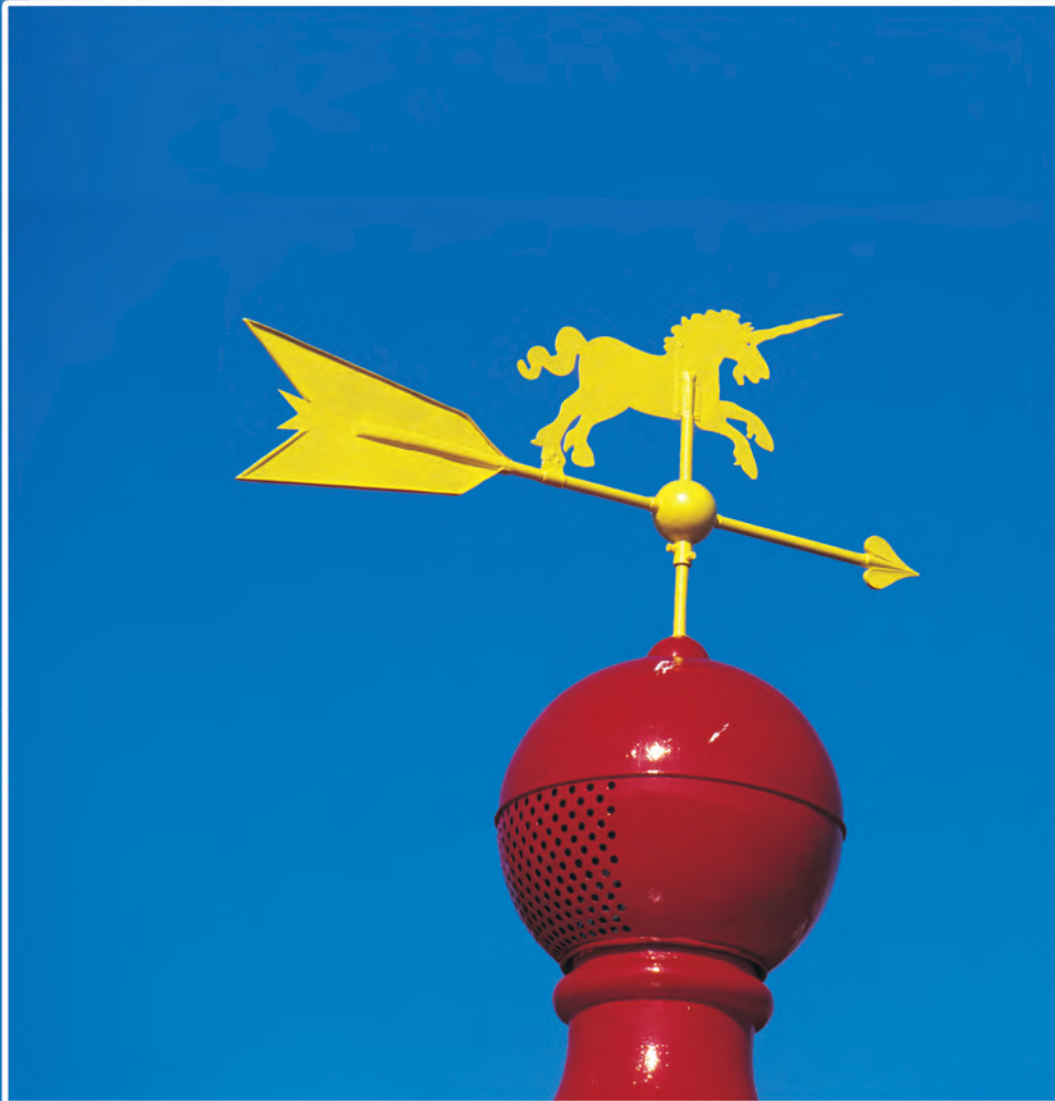


The vector representation that we use today is attributed to Josiah Willard Gibbs (1839-1903). Gibbs was born in the town of New Haven in Connecticut (his father had also been a professor at Yale), and his main scientific accomplishments were in physics. He read Maxwell’s *Treatise on Electricity and Magnetism*, and also studied Grassmann’s *The Calculus of Extension*. He concluded that vectors would provide a more efficient tool for his work in physics. In 1881, Gibbs privately printed notes on vector analysis for his students, which were widely distributed to scholars in the United States, Britain, and Europe. The first book on modern vector analysis in English was *Vector Analysis* (1901), which was Gibbs’s notes as assembled by one of his last graduate students, Edwin B. Wilson (1879-1964). Another contribution to the modern understanding and use of vectors was made by Jean Frenet (1816-1990).



Vector methods were introduced in Italy (1887, 1888, 1897), Russia (1907), and the Netherlands (1903). Vectors are now the modern language of a great deal of physics and applied mathematics, and they continue to hold their own mathematical interest.

VECTORS IN THE PLANE



A. BASIC VECTOR CONCEPTS

Some of the quantities we measure in our daily lives are completely determined by their magnitudes, for example, length, mass, area, temperature, and energy. When we speak of a length of 3 cm or an area of 5 cm², we only need one number to describe each of these quantities. We call such quantities **scalar quantities**.

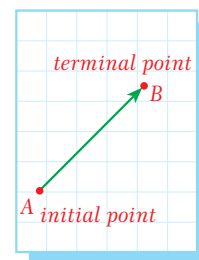
On the other hand, to describe a force, we need to record its direction as well as its size. For example, to describe the velocity of a moving object, we must specify both the speed and the direction of travel. Quantities such as displacement, velocity, acceleration, and other forces that have magnitude as well as direction are called **vector quantities**. We usually show a vector quantity as an arrow that points in the direction of the action, with length that shows the magnitude of the action in terms of a suitable unit. The way to represent such quantities mathematically is through the use of **vectors**.

1. Directed Line Segment

When we move from Antalya to Berlin by bus, we have two quantities: the direction from Antalya to Berlin, and the length of the displacement between these cities.



We can sketch a line segment AB as shown in the figure with starting point A and finishing point B to represent the movement from Antalya to Berlin. The line segment AB with an arrow has direction and length. The arrow head specifies the direction, and the length of the arrow specifies the magnitude, at a suitable scale. A and B are the endpoints of the segment. Point A is called the **initial point** and point B is called the **terminal point** of the line segment. The resulting segment AB is called a **directed line segment**.



Definition

directed line segment

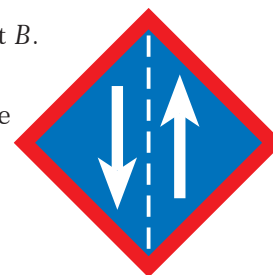
A line segment with direction is called a **directed line segment**.

We write $\vec{AB}$ to denote a directed line segment from point A to point B .



Directed line segments are used in daily life. For example, some traffic signs for drivers use directed line segments.

In technology we also use directed line segments.



EXAMPLE

1 Points M, N, P and K on line d are given. Write all the directed line segments with endpoints M, N, P , or K .



Solution The directed line segments with endpoints M, N, P , or K are $\overrightarrow{MN}, \overrightarrow{MP}, \overrightarrow{MK}, \overrightarrow{NP}, \overrightarrow{NK}, \overrightarrow{NM}, \overrightarrow{PK}, \overrightarrow{PN}, \overrightarrow{PM}, \overrightarrow{KP}, \overrightarrow{KN}$, and $\overrightarrow{KM}$.

Notice that $\overrightarrow{MN}$ is not the same as $\overrightarrow{NM}$, and $\overrightarrow{MP}$ is not the same as $\overrightarrow{PM}$. This is because the line segments have direction. Pairs such as $\overrightarrow{MN}$ and $\overrightarrow{NM}$ have the same magnitude but opposite direction.

2. Definition of a Vector

Definition

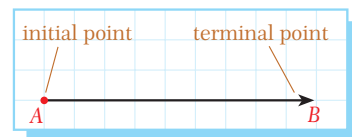
vector

A directed line segment in the plane is called a **vector**.

The length of the directed line segment is the length of the vector.

The direction of the directed line segment is the direction of the vector.

We write $\overrightarrow{AB}$ to mean a vector with initial point A and terminal point B . Alternatively, we can name a vector with a lower-case letter such as $\vec{u}$ or $\vec{p}$.



For example, consider a line segment $\overrightarrow{AB}$ with length 2 cm.

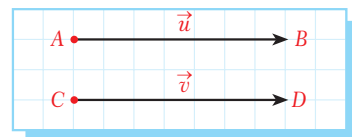
We can say the length of vector $\overrightarrow{AB}$ is 2 cm, and write $|\overrightarrow{AB}| = 2 \text{ cm}$.

3. Equal Vectors

Definition

equal vectors

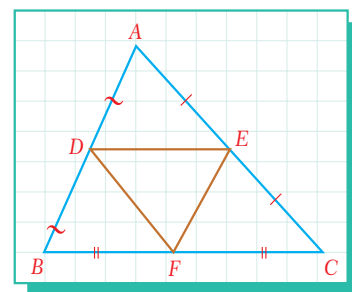
Two vectors that have the same direction and length are called **equal vectors**. We show that two vectors $\vec{u}$ and $\vec{v}$ are equal by writing $\vec{u} = \vec{v}$.



EXAMPLE

2 In the figure, D, E , and F are the midpoints of AB, AC and BC respectively, and $DE \parallel BC, EF \parallel AB, DF \parallel AC$. Name all the equal vectors.

Solution In triangle ABC , $|DE| = |BF| = |FC|$
 $|EF| = |AD| = |DB|$
 $|DF| = |AE| = |EC|$.
 So $\overrightarrow{DE} = \overrightarrow{BF} = \overrightarrow{FC}$ $\overrightarrow{ED} = \overrightarrow{FB} = \overrightarrow{CF}$
 $\overrightarrow{EF} = \overrightarrow{AD} = \overrightarrow{DB}$ and $\overrightarrow{FE} = \overrightarrow{DA} = \overrightarrow{BD}$
 $\overrightarrow{DF} = \overrightarrow{AE} = \overrightarrow{EC}$ $\overrightarrow{FD} = \overrightarrow{EA} = \overrightarrow{CE}$.



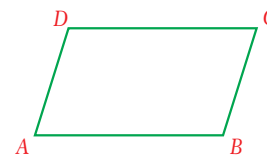
Check Yourself 1

$ABCD$ is a parallelogram in the figure.

- How many pairs of equivalent directed line segments are there?
- How many pairs of equal vectors are there?

Answers

- 4
- 4

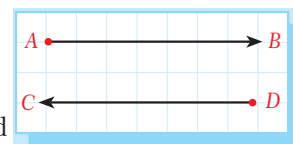


Definition

opposite vectors

Two vectors are called **opposite vectors** if and only if their magnitudes (lengths) are the same but their directions are opposite.

For example, in the figure, $\vec{AB}$ and $\vec{BA}$ are opposite vectors. $\vec{CD}$ and $\vec{DC}$ are also opposite vectors. We can write $\vec{AB} = -\vec{BA}$ and $\vec{CD} = -\vec{DC}$.



Definition

zero vector

A vector whose initial and terminal points are the same is called a **zero vector**.

We write a zero vector as $\vec{0}$.

A zero vector has no direction and no size.

B. VECTOR OPERATIONS

1. Addition of Vectors

Let $\vec{PQ}$ and $\vec{QR}$ be two vectors in a plane. $\vec{PQ} + \vec{QR}$ denotes the **sum** of the vectors $\vec{PQ}$ and $\vec{QR}$. There are two ways to find the sum of two or more vectors.

a. The Polygon Method

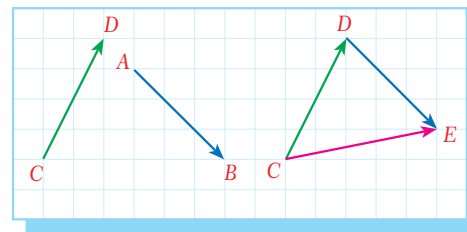
Imagine we want to add n vectors together. Using the polygon method, we draw the first vector. Then we place the initial point of the second vector at the terminal point of the first vector, the initial point of the third vector at the terminal point of the second vector, and so on until we place the initial point of the n^{th} vector at the terminal point of the $(n - 1)^{\text{th}}$ vector. The sum is the vector whose initial point is the initial point of the first vector and whose terminal point is the terminal point of the last vector.

Let us look at an example.

Let $\vec{AB}$ and $\vec{CD}$ be two vectors in a plane, as in the diagram. We place the initial point of $\vec{AB}$ at the terminal point of $\vec{CD}$ to make $\vec{DE}$ ($\vec{AB} = \vec{DE}$).

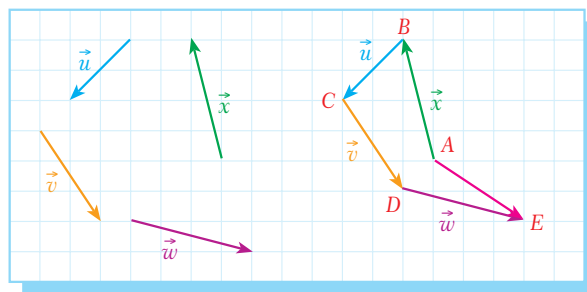
Using the polygon method,

$$\vec{CD} + \vec{AB} = \vec{CD} + \vec{DE} = \vec{CE}.$$



Now look at an example of adding more than two vectors using the polygon method.
As shown in the figure,

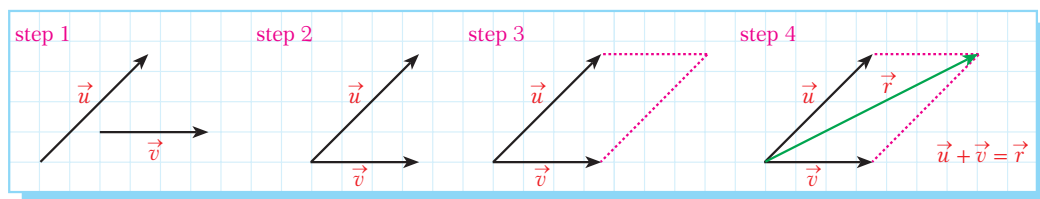
$$\vec{u} + \vec{v} + \vec{w} + \vec{x} = \vec{AE}.$$



b. The Parallelogram Method

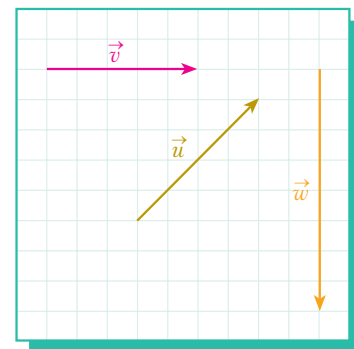
To add one vector to another using the parallelogram method, we draw the first vector, and then we draw the second vector with its initial point at the initial point of the first vector. We make a parallelogram by drawing two additional sides, each passing through the terminal point of one of the vectors and parallel to the other vector. We find the sum by drawing a vector along the diagonal from the common initial point to the intersection of the two lines.

Look at the example of adding $\vec{u}$ and $\vec{v}$ using the parallelogram method:

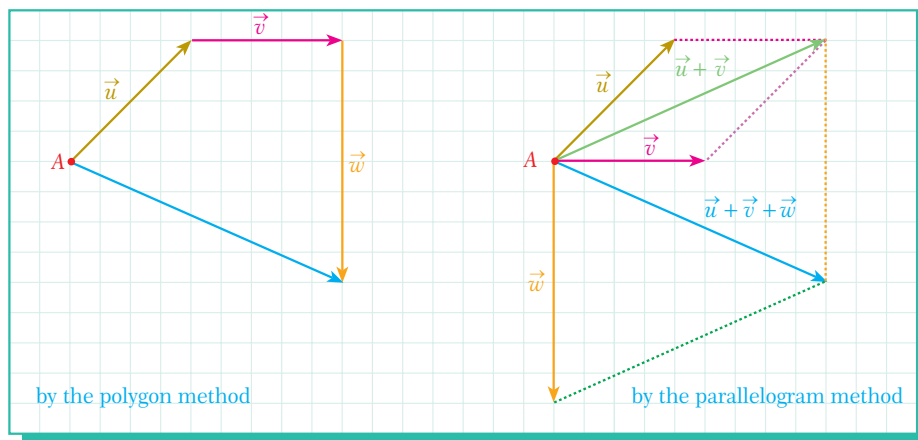


EXAMPLE

3 Find $\vec{u} + \vec{v} + \vec{w}$ in the figure on the right.



Solution Let us choose A as a fixed point. We can use the polygon method or the parallelogram method to add the given vectors $\vec{u} + \vec{v} + \vec{w}$.



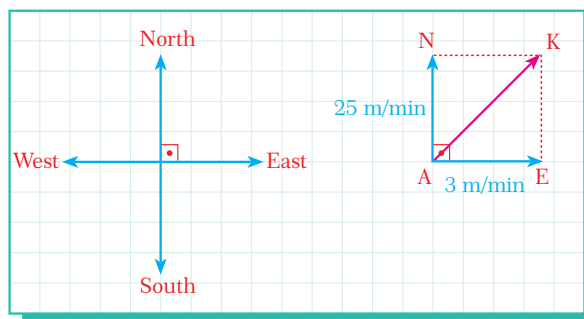
EXAMPLE

- 4** The velocity of a boat is 25 m/min north and the velocity of a river current is 3 m/min east. Draw a scale diagram to show the velocities as vectors and find the sum.

Solution First we choose a starting point A and write $\vec{AN}$ = velocity of the boat due north.
 $\vec{AE}$ = velocity of the current due east.
 $\vec{AN}$ and $\vec{AE}$ are perpendicular, and $\vec{AK}$ is the sum of $\vec{AN}$ and $\vec{AE}$:

$$|\vec{AK}| = \sqrt{25^2 + 3^2} = \sqrt{634}.$$

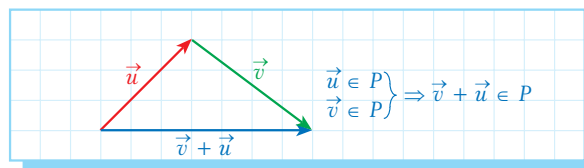
This is the sum of the vectors.



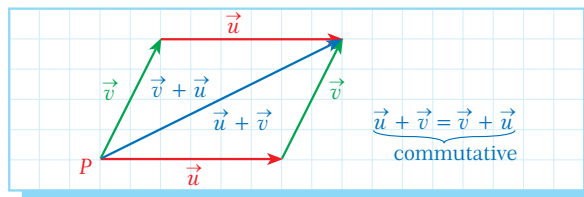
c. Properties of Vector Addition

Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be three vectors in a plane P .

1. The sum of any two vectors in P is also a vector in P (closure property).

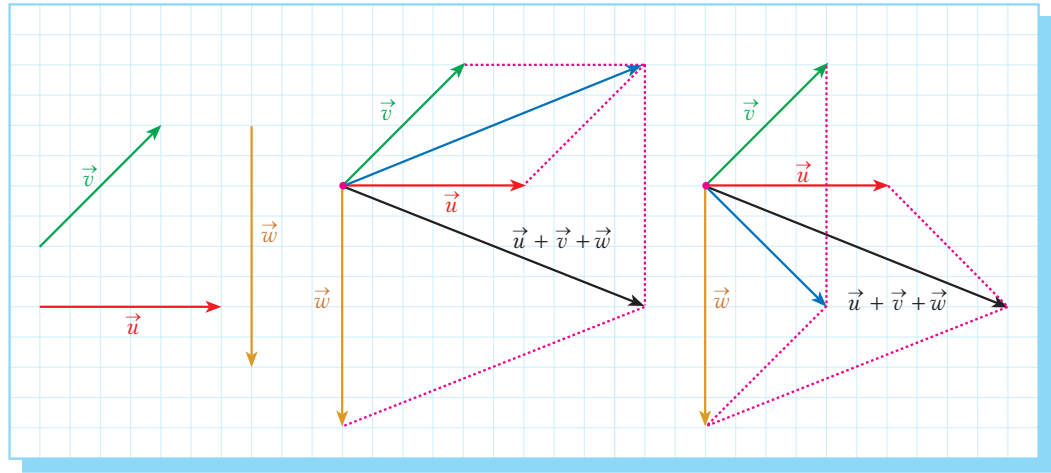


2. The sum of any two vectors in P is commutative (commutative property).

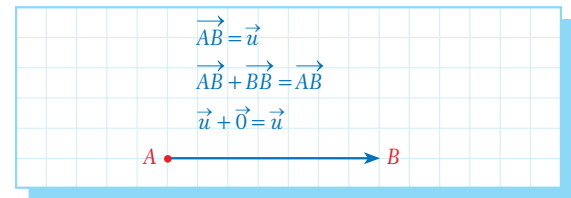


3. The sum of any three vectors in P is associative (associative property).

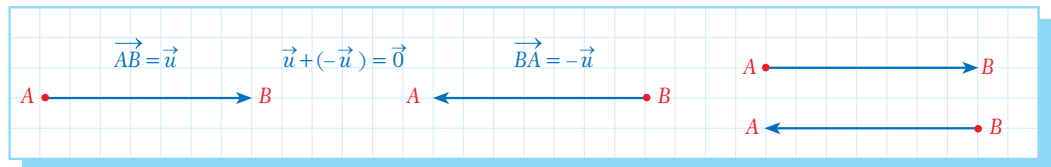
$$\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$$



4. The sum of the zero vector and a vector in P is the vector itself (identity element).



5. The additive inverse of any vector $\vec{u}$ is $-\vec{u}$: $\vec{u} + (-\vec{u}) = \vec{0}$ (additive inverse).

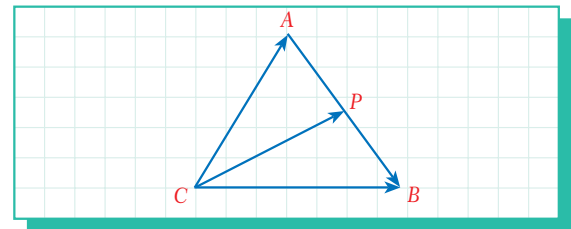


EXAMPLE

- 5 In a triangle ABC , P is the midpoint of $\overline{AB}$. Express $\overrightarrow{CP}$ in terms of $\overrightarrow{CA}$ and $\overrightarrow{CB}$.

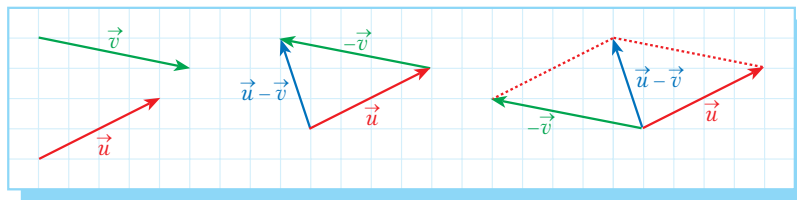
Solution

$$\begin{aligned} \overrightarrow{CP} &= \overrightarrow{CA} + \overrightarrow{AP} \\ + \quad \overrightarrow{CP} &= \overrightarrow{CB} + \overrightarrow{BP} \\ \hline 2 \cdot \overrightarrow{CP} &= \overrightarrow{CA} + \overrightarrow{CB} + \underbrace{\overrightarrow{AP} + \overrightarrow{BP}}_{\vec{0}} \\ \overrightarrow{CP} &= \frac{1}{2} \cdot (\overrightarrow{CA} + \overrightarrow{CB}) \end{aligned}$$



2. Subtraction of Vectors

Since subtraction is the inverse of addition, we can find the difference of two vectors $\vec{u}$ and $\vec{v}$ by adding the vectors $\vec{u}$ and $-\vec{v}$ (opposite of $\vec{v}$) using either the parallelogram method ($\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$) or the polygon method.



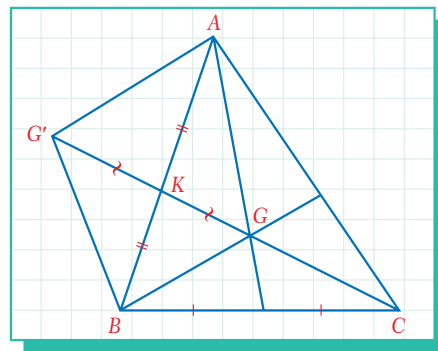
EXAMPLE

6 In a triangle ABC , G is the centroid. Find $\vec{GA} + \vec{GB} + \vec{GC}$.

Solution Let us label a point G' on the extension of CG which satisfies $|CG| = |GG'|$. Since G is the centroid of $\triangle ABC$, $|CG| = 2 \cdot |GK|$. Therefore $|GG'| = 2 \cdot |GK|$, which means that K is the midpoint of GG' . We conclude that $AG'BG$ is a parallelogram because K is the midpoint of both diagonals AB and GG' . So we have $\vec{AG'} = \vec{GB}$ which gives us $\vec{GA} + \vec{GB} = \vec{GG'}$.

On the other hand, we have $\vec{CG} = \vec{GG'} = -\vec{GC}$. Using this result in $\vec{GA} + \vec{GB} = \vec{GG'}$, we get $\vec{GA} + \vec{GB} = -\vec{GC}$ which gives us

$$\vec{GA} + \vec{GB} + \vec{GC} = \vec{0}.$$

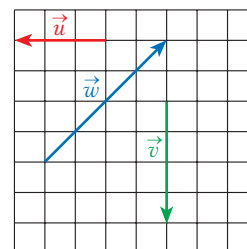


Check Yourself 2

- Find the following using the vectors in the figure.
 - $\vec{v} - \vec{u}$
 - $\vec{u} + \vec{w}$
 - $\vec{w} + \vec{v} - \vec{u}$
- In a triangle ABC , $D \in [BC]$ and $|BD| = 2 \cdot |DC|$. Express $\vec{AD}$ in terms of $\vec{AB}$ and $\vec{AC}$.

Answers

- use the polygon method
- $\vec{AD} = \frac{2}{3}\vec{AC} + \frac{1}{3}\vec{AB}$



3. Multiplication of a Vector by a Scalar

Multiplying a vector by a scalar makes the vector longer or shorter depending on the value of the scalar. If the scalar is greater than 1 or less than -1 , multiplying makes a longer vector. If the scalar is between -1 and 1 and non-zero, it makes a shorter vector.

If the scalar is positive, multiplying does not change the direction.

If the scalar is negative, multiplying will make the vector's direction opposite.

For a real number a and a vector $\vec{u}$,

1. if $a > 0$ then vector $a \cdot \vec{u}$ has the same direction as $\vec{u}$ and the length $|a \cdot \vec{u}| = a \cdot |\vec{u}|$.
2. if $a < 0$ then vector $a \cdot \vec{u}$ has the opposite direction to $\vec{u}$ and the length $|a \cdot \vec{u}| = |a| \cdot |\vec{u}|$.
3. if $a = 0$ then $a \cdot \vec{u} = \vec{0}$.

EXAMPLE

7

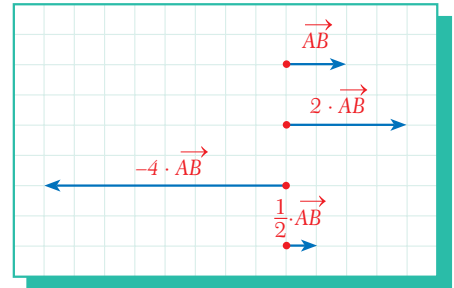
Using $\vec{AB}$ as shown in the figure, draw vector diagrams to show $2 \cdot \vec{AB}$, $-4 \cdot \vec{AB}$, and $\frac{1}{2} \cdot \vec{AB}$.



Solution

Since 2 and $\frac{1}{2}$ are positive, $2 \cdot \vec{AB}$ and $\frac{1}{2} \cdot \vec{AB}$ have the same direction as $\vec{AB}$. However, $2 \cdot \vec{AB}$ is twice as long as $\vec{AB}$ and $\frac{1}{2} \cdot \vec{AB}$ is half as long.

On the other hand, $-4 \cdot \vec{AB}$ has opposite direction to $\vec{AB}$ (since -4 is a negative scalar) and it is four times as long as $\vec{AB}$.



a. Properties of the Multiplication of a Vector by a Scalar

For any vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ and real numbers a and b , the following properties are satisfied.

1. $a \cdot \vec{u}$ is a vector in the plane
2. $(a \cdot b) \cdot \vec{u} = a \cdot (b \cdot \vec{u})$
3. $(a + b) \cdot \vec{u} = a \cdot \vec{u} + b \cdot \vec{u}$
4. $a \cdot (\vec{u} + \vec{v}) = a \cdot \vec{u} + a \cdot \vec{v}$
5. $1 \cdot \vec{u} = \vec{u}$
6. $a \cdot \vec{0} = \vec{0}$

EXAMPLE

8 Points A, B, C , and M are on the same line. M is between C and B . $\vec{AB} = 3\vec{AC}$. Express the vector $\vec{MC}$ in terms of vectors $\vec{MA}$ and $\vec{MB}$.

Solution

$$\vec{AB} = 3\vec{AC} \text{ so } \vec{CB} = 2\vec{AC} \quad (1)$$

$$\vec{MA} + \vec{AC} = \vec{MC} \quad (2)$$

$$\vec{CM} + \vec{MB} = 2\vec{AC} \quad (3)$$

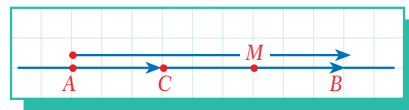
$$\vec{AC} = -\frac{1}{2}\vec{MC} + \frac{1}{2}\vec{MB} \quad (4)$$

$$\vec{MA} - \frac{1}{2}\vec{MC} + \frac{1}{2}\vec{MB} = \vec{MC} \text{ by (2) and (4).}$$

$$\vec{MA} + \frac{1}{2}\vec{MB} = \frac{1}{2}\vec{MC} + \vec{MC}$$

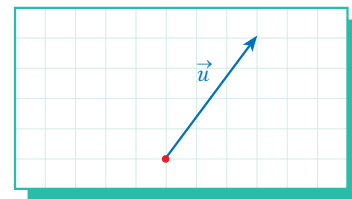
$$\vec{MA} + \frac{1}{2}\vec{MB} = \frac{3}{2}\vec{MC}$$

$$\text{So } \vec{MC} = \frac{2}{3}\vec{MA} + \frac{1}{3}\vec{MB}.$$



Check Yourself 3

1. Multiply the vector $\vec{u}$ by the scalars -2 , 3 , 0.5 and $-\frac{1}{3}$ and draw a vector diagram to show them.
2. Points A, B, C , and M are on the same line. M is between A and C . $\vec{AB} = 2\vec{AC}$. Express the vector $\vec{MC}$ in terms of the vectors $\vec{MA}$ and $\vec{MB}$.



Answers

$$2. \vec{MC} = \frac{1}{2}(\vec{MA} + \vec{MB})$$

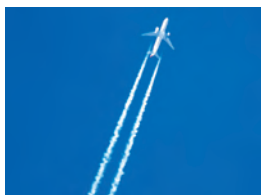
C. PARALLEL VECTORS

1. Parallel Vectors

Definition

parallel vectors

Let $\vec{a}$ and $\vec{b}$ be two vectors. $\vec{a}$ and $\vec{b}$ are called **parallel vectors** if and only if $\vec{a} = k\vec{b}$ where $k \neq 0$ and $k \in \mathbb{R}$. We write $\vec{a} \parallel \vec{b}$ to show that two vectors are parallel.

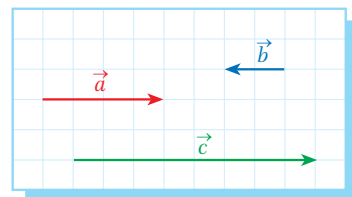


For example, in the diagram, $|\vec{a}| = 2$ cm,

$|\vec{b}| = 1$ cm and $|\vec{c}| = 4$ cm.

We can express vector $\vec{a}$ as $\vec{a} = \frac{1}{2}\vec{c}$ and $\vec{a} = -2\vec{b}$.

Therefore the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are parallel, i.e. $\vec{a} \parallel \vec{b} \parallel \vec{c}$.



EXAMPLE**9**

In a triangle ABC , D and E are the midpoints of sides AB and AC respectively.

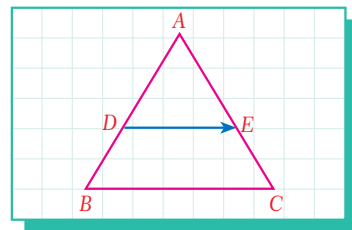
Show that $\overrightarrow{DE} \parallel \overrightarrow{BC}$.

Solution $\overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BC}$ and $\overrightarrow{DA} + \overrightarrow{AE} = \overrightarrow{DE}$ by the addition of vectors.

$$\overrightarrow{DA} = \frac{1}{2} \cdot \overrightarrow{BA}, \quad \overrightarrow{AE} = \frac{1}{2} \cdot \overrightarrow{AC}$$

$$\overrightarrow{DE} = \frac{1}{2} \cdot \overrightarrow{BA} + \frac{1}{2} \cdot \overrightarrow{AC} = \frac{1}{2} \cdot (\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2} \cdot \overrightarrow{BC}$$

Now $\overrightarrow{DE} = \frac{1}{2} \cdot \overrightarrow{BC}$, so $\overrightarrow{DE} \parallel \overrightarrow{BC}$ by the definition of parallel vectors.

**EXAMPLE****10**

In a quadrilateral $ABCD$, points E and F are the midpoints of side AB and diagonal AC , respectively. Show that $EF \parallel BC$.

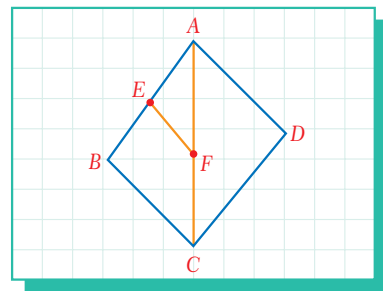
Solution $\overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BC}$ (1) and $\overrightarrow{EA} + \overrightarrow{AF} = \overrightarrow{EF}$ (2).

$$\overrightarrow{EA} = \frac{1}{2} \cdot \overrightarrow{BA}$$

$$+ \overrightarrow{AF} = \frac{1}{2} \cdot \overrightarrow{AC}$$

$$\overrightarrow{EA} + \overrightarrow{AF} = \frac{1}{2} \cdot (\overrightarrow{BA} + \overrightarrow{AC})$$

$$\overrightarrow{EF} = \frac{1}{2} \cdot \overrightarrow{BC} \text{ by (1) and (2). Therefore, } \overrightarrow{EF} \parallel \overrightarrow{BC}.$$



2. Non-Parallel Vectors

By the definition of parallel vectors we can conclude that if $\vec{a}$ and $\vec{b}$ are non-zero, non-parallel vectors, then $h \cdot \vec{a} = k \cdot \vec{b}$ when $h = k = 0$. Look at the proof:

Suppose that $h \neq k \neq 0$, $\vec{a} = \frac{k}{h} \cdot \vec{b}$.

Then $\vec{a} \parallel \vec{b}$. This is a contradiction, since $\vec{a}$ and $\vec{b}$ are non-parallel. As a result, $h = k = 0$.

EXAMPLE**11**

Prove that the diagonals of a parallelogram intersect at their midpoints by using vectors.

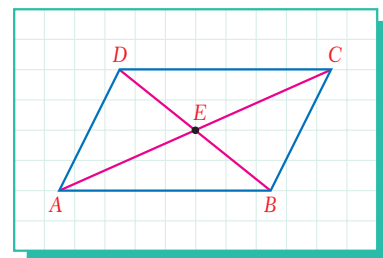
Solution Look at the diagram. Let $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BC} = \vec{b}$, so

$$\overrightarrow{AC} = \vec{a} + \vec{b}$$

$$\overrightarrow{DB} = \vec{a} - \vec{b}$$

$$\overrightarrow{AE} = m \cdot (\vec{a} + \vec{b})$$

$$\overrightarrow{EB} = n \cdot (\vec{a} - \vec{b})$$



$$\begin{aligned}\vec{AE} + \vec{EB} &= m \cdot (\vec{a} + \vec{b}) + n \cdot (\vec{a} - \vec{b}) \\ \vec{a} &= m \cdot (\vec{a} + \vec{b}) + n \cdot (\vec{a} - \vec{b}) \\ (m + n - 1) \cdot \vec{a} &= \vec{b} \cdot (n - m).\end{aligned}$$

Since $\vec{a}$ and $\vec{b}$ are non-zero and non-parallel, we have $(m + n - 1) = (n - m) = 0$. Therefore $m + n = 1$ and $m = n$, and so $m = n = \frac{1}{2}$.

EXAMPLE

12

In a triangle ABC , D , E and F are the midpoints of sides BC , BA and AC respectively. Find the sum $\vec{AD} + \vec{BF} + \vec{CE}$.

Solution

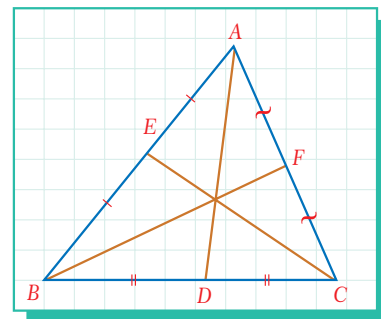
Let us draw the triangle ABC as in the figure.

If D , E and F are midpoints then

$$\begin{aligned}\vec{AD} &= \frac{1}{2} \cdot (\vec{AB} + \vec{AC}) \quad \text{by the result of Example 5.} \\ \vec{BF} &= \frac{1}{2} \cdot (\vec{BA} + \vec{BC}) \\ + \vec{CE} &= \frac{1}{2} \cdot (\vec{CA} + \vec{CB})\end{aligned}$$

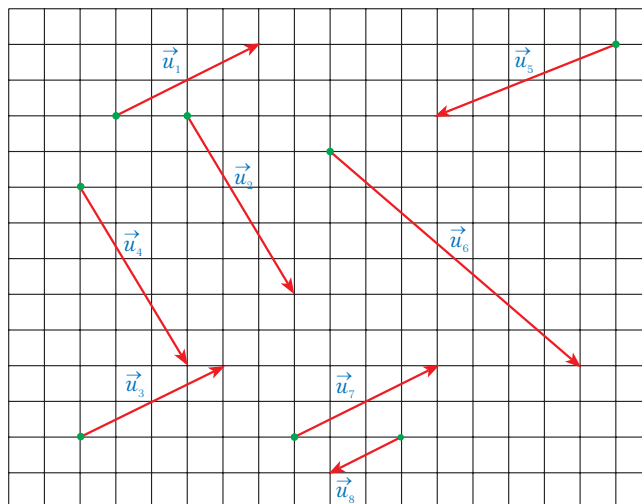
$$\vec{AD} + \vec{BF} + \vec{CE} = \frac{1}{2} \cdot (\underbrace{\vec{AB} + \vec{BA}}_{\vec{0}} + \underbrace{\vec{AC} + \vec{CA}}_{\vec{0}} + \underbrace{\vec{BC} + \vec{CB}}_{\vec{0}})$$

Therefore, $\vec{AD} + \vec{BF} + \vec{CE} = \vec{0}$.



Check Yourself 4

1. Name all the pairs of parallel vectors in the figure.
2. Add the vector pairs $\vec{u}_1$ and $\vec{u}_5$, $\vec{u}_2$ and $\vec{u}_6$, and $\vec{u}_3$ and $\vec{u}_7$.
3. Find the additive inverse of $\vec{u}_7$ and $\vec{u}_1$.
4. Subtract $\vec{u}_8$ from $\vec{u}_4$.
5. In a quadrilateral $ABCD$, P and K are the midpoints of sides AB and DC respectively. Express $\vec{PK}$ in terms of $\vec{DA}$ and $\vec{CB}$.



Answers

1. look at the directions and lengths
2. use the polygon method
3. use the polygon method
4. use the polygon method
5. $\vec{PK} = -\frac{1}{2} \cdot (\vec{DA} + \vec{CB})$

Traffic Signs



TWO WAY
TRAFFIC



Traffic signs are important for drivers and pedestrians. If people do not know the meaning of these signs, they can have some difficulties in traffic.

For example, if a driver drives in the opposite direction to a 'one way' sign, he or she might have an accident.

Project: Describe some other areas in which we use directed line segments (for example: flowcharts, keyboards, ...).

EXERCISES 1

A. Basic Vector Concepts

1. Draw any two parallel directed line segments with the same length but opposite direction. Express one of the line segments in terms of the other.
2. How many equal directed line segments can we find on two parallel lines?
3. Make a scale diagram showing the vectors in each statement and find their sum.
 - a. A 6 km trip east is followed by a 3 km trip southeast.
 - b. The velocity of a swimmer is 5 m/min west and the velocity of a river current is 2 m/min north.

B. Vector Operations

4. Using the vectors given on the right, sketch the following vectors.

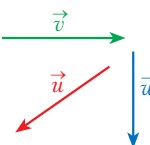
a. $\vec{u} + \vec{v}$

b. $\vec{w} - (\vec{u} + \vec{v})$

c. $-\vec{w} - \vec{v} - \vec{u}$

d. $\vec{u} + (\vec{w} + \vec{v})$

e. $\vec{u} - 3 \cdot \vec{v} + 2 \cdot \vec{w}$



5. In a plane, $[AB]$ is given. Point K is the midpoint of $[AB]$ and point O is any point in the same plane. Express $\vec{OK}$ in terms of $\vec{OA}$ and $\vec{OB}$.
6. In a triangle ABC , points D and E lie on $[BC]$ and $|BD| = |DE| = |EC|$. Express the vector $\vec{AD} + \vec{AE}$ in terms of $\vec{AB}$ and $\vec{AC}$.

7. Show that $(\vec{AB} + \vec{BC}) + \vec{CD} = \vec{AB} + (\vec{BC} + \vec{CD})$ by using the parallelogram method.

8. Point O is in the plane of a triangle ABC . Point G is the centroid of triangle ABC . Show that $\vec{OA} + \vec{OB} + \vec{OC} = 3\vec{OG}$.

9. Consider any two points A and B in a plane. For any point P in the same plane, the symmetry of point P with respect to point A is Q and the symmetry of point Q with respect to point B is point R . Show that $\vec{PR}$ is always $2\vec{AB}$. (Hint: let A , B , and C be collinear. If $|AB| = |BC|$ then A is the symmetry of C with respect to B .)

C. Parallel Vectors

10. In a triangle ABC , G is the point of intersection of the medians and P is the midpoint of $\vec{BG}$. Show that $\vec{PA} + \vec{PC} = 4\vec{PG}$.

11. $ABCD$ is a quadrilateral and M, N, P, Q are the midpoints of AB, BC, CD , and DA respectively. Show that $MNPQ$ is a parallelogram.

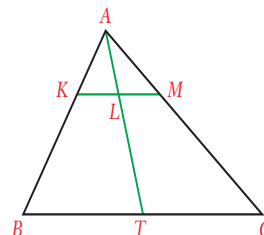
12. In a six-sided polygon $ABCDEF$, $\vec{AB} = \vec{ED}$, $\vec{BC} = \vec{FE}$, and $\vec{CD} = \vec{AF}$. Show that $FBCE$ is a parallelogram.

13. A trapezoid is a four-sided figure with only two parallel sides. A line segment which joins the midpoints of the non-parallel sides is called the *median* of the trapezoid. Prove that the median of a trapezoid is parallel to the two parallel sides, and has magnitude equal to half of their sum.

14. In a triangle ABC , $|BD| = |DE| = |EC|$, and $E, D \in [BC]$. If $|\vec{AD} + \vec{AE}| = 9$ cm, find $|\vec{AB} + \vec{AC}|$.

15. In a quadrilateral $ABCD$, E and F are the midpoints of the diagonals AC and BD respectively. Show that $\vec{AB} + \vec{AD} + \vec{CB} + \vec{CD} = 4\vec{EF}$.

16. In the figure,
 $\star$ T is the midpoint of BC ,
 ABC is a triangle,
 $2 \cdot |AK| = |KB|$, and
 $2 \cdot |AM| = |MC|$.
 Use vectors to show that
 $|\vec{AL}| = \frac{1}{2} \cdot |\vec{LT}|$.



17. Show that the centroid of a triangle divides a median in the ratio 1:2 using vectors.
 $\star\star$

18. In a parallelogram $ABCD$, K is the midpoint of side DC , and the intersection point of diagonals AC and BK is T . Show that $\vec{AC} = 3\vec{TC}$.
 $\star\star$



ANALYSIS OF VECTORS ANALYTICALLY

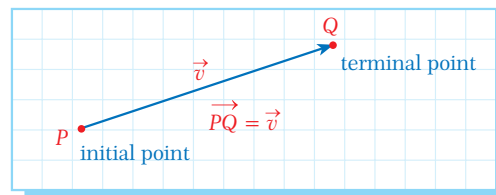
We have studied vectors geometrically. Now let us look at a method for describing vectors analytically.

We will begin this section by looking at some important axioms.

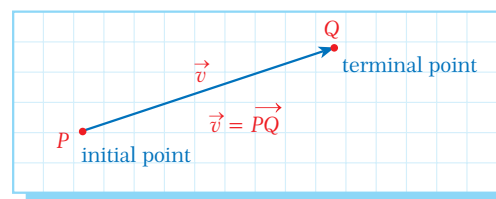
A. BASIC CONCEPTS OF VECTORS IN THE ANALYTIC PLANE

1. Axioms

1. For each pair of points P and Q there exists a unique vector $\vec{v}$ such that $\vec{PQ} = \vec{v}$.



2. For each point P and vector $\vec{v}$, there is a unique point Q such that $\vec{v} = \vec{PQ}$.



Conclusion

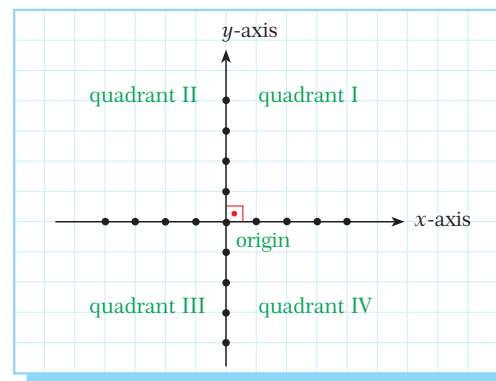
1. Two points in a plane determine two opposite vectors.
2. In a plane, if one point is fixed as an initial point then all the other points in the plane can be chosen as the terminal point of any vector.

2. The Rectangular Coordinate System

The **rectangular coordinate system** is formed by two perpendicular intersecting number lines, as shown in the diagram opposite.

1. The horizontal number line is called the **x-axis**.
2. The vertical number line is called the **y-axis**.

The **origin** is the point of intersection. At this intersection, both number lines are 0. The rectangular coordinate system is split into four **quadrants**, which are marked in the diagram with roman numerals.



Each point in the coordinate system is associated with a **pair of real numbers**. In an x, y system, the **x-coordinate** always comes first and the **y-coordinate** always comes second in the pair (x, y) . The first coordinate is called the **abscissa** of the point and the second coordinate is called the **ordinate** of the point.

EXAMPLE

13

Plot each pair of coordinates and name the quadrant or axis in which the point lies.

$A(2, 3)$, $B(-1, 2)$, $C(-3, -4)$, $D(2, 0)$, $E(0, 5)$

Solution

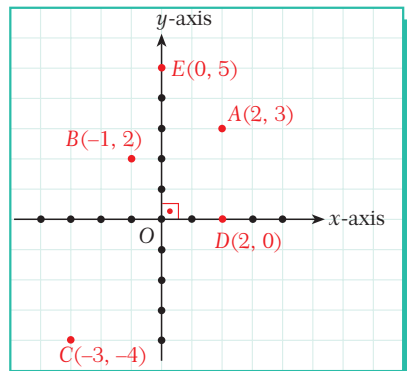
$A(2, 3)$ lies in quadrant I.

$B(-1, 2)$ lies in quadrant II.

$C(-3, -4)$ lies in quadrant III.

$D(2, 0)$ lies on the x -axis.

$E(0, 5)$ lies on the y -axis.



Check Yourself 5

Plot the pairs of real numbers and name the quadrant or axis in which the point lies.

$A(3, 2)$, $B(-2, 1)$, $C(-4, -3)$, $D(0, 2)$, $E(5, 0)$

3. Position Vector

Definition

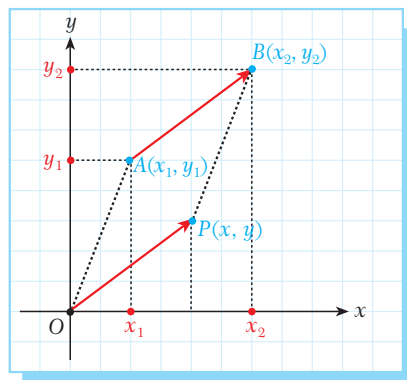
position vector

A vector $\vec{OP}$ whose initial point is at the origin of the rectangular coordinate plane and which is parallel to a vector $\vec{AB}$ is called the **position vector** of $\vec{AB}$ in the plane. In other words, if $\vec{OP}$ is the position vector of $\vec{AB}$, then $\vec{OP} \parallel \vec{AB}$, $|\vec{OP}| = |\vec{AB}|$, and $\vec{OP} = \vec{AB}$.

From this definition we can conclude the following:

- For every vector in the plane there exists a position vector $\vec{OP}$ which is determined by a pair (x, y) , that is, $\vec{OP} = (x, y)$.
- If $\vec{OP} = \vec{AB}$, then $OPBA$ is a parallelogram.

Look at the diagram. We can calculate that $x = x_2 - x_1$ and $y = y_2 - y_1$. Therefore the vector $\vec{AB}$ determined by the points $A(x_1, y_1)$ and $B(x_2, y_2)$ has position vector $\vec{OP} = \vec{AB} = \vec{OB} - \vec{OA} = (x, y) = (x_2 - x_1, y_2 - y_1)$.



EXAMPLE

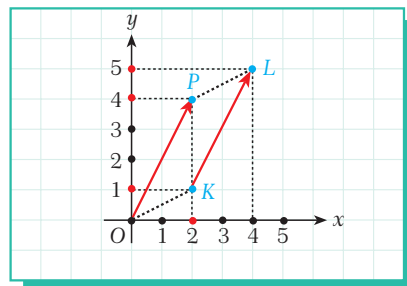
14

Find the position vector of $\vec{KL}$ with endpoints $K(2, 1)$ and $L(4, 5)$.

Solution

$$\vec{OP} = \vec{KL} = \vec{OL} - \vec{OK} = (4 - 2, 5 - 1)$$

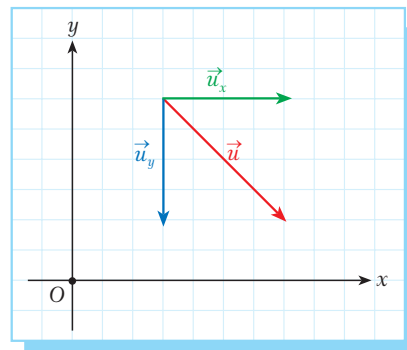
$$\vec{OP} = (2, 4)$$



4. Components of a Vector

We have seen how to describe a vector in the plane with reference to its unique position vector.

We can also express a vector in the plane as the sum of two vectors, one of which is parallel to the x -axis and the other parallel to the y -axis. These two vectors are called the **components** of the original vector. The component parallel to the x -axis is called the **horizontal component** of the vector and the component parallel to y -axis is called the **vertical component**. Expressing a vector as the sum of its components like this is called **resolving** the vector. For example, in the figure



We can represent the vector $\vec{u}$ as an ordered pair of real numbers:

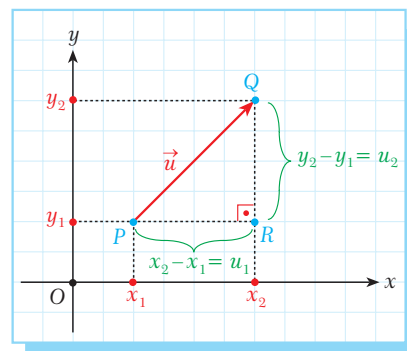
$$\vec{u} = (u_1, u_2) \text{ or } \vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix},$$

where u_1 is the horizontal scalar component of $\vec{u}$, and u_2 is the vertical scalar component of $\vec{u}$.

Now, $\vec{u}_x = (u_1, 0)$ and $\vec{u}_y = (0, u_2)$.

Look at the diagram opposite. By applying the Pythagorean theorem for triangle PQR in the figure, we can see that the length of the vector $\vec{u} = (u_1, u_2)$ is

$$|\vec{u}| = \sqrt{u_1^2 + u_2^2}.$$



EXAMPLE

15

Find the length of each vector.

a. $\vec{u} = (2, -3)$

b. $\vec{v} = (3, 0)$

c. $\vec{w} = \left(\frac{3}{5}, \frac{4}{5}\right)$

Solution a. $|\vec{u}| = \sqrt{2^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13}$

b. $|\vec{v}| = \sqrt{3^2 + 0^2} = \sqrt{9} = 3$

c. $|\vec{w}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{9+16}{25}} = \sqrt{\frac{25}{25}} = 1$

EXAMPLE

16

Find the length of the vector $\vec{u}$ with initial point $(1, 2)$ and terminal point $(3, 6)$.

Solution $\vec{u} = (3 - 1, 6 - 2) = (2, 4)$

$$|\vec{u}| = \sqrt{2^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20}$$

EXAMPLE

17

Find the possible values of a given $|\vec{AB}| = 2\sqrt{5}$ and the endpoints $A(a, 2)$ and $B(1, 4)$.

Solution

$$\begin{aligned} |\vec{AB}| &= \sqrt{(1-a)^2 + (4-2)^2} = \sqrt{(1-a)^2 + 4} = 2\sqrt{5} \\ (1-a)^2 + 4 &= 20 \\ (1-a)^2 &= 16 \\ 1-a &= 4 \quad \text{or} \quad 1-a = -4 \\ a &= -3 \quad \text{or} \quad a = 5 \end{aligned}$$

Check Yourself 6

1. Find the position vector of $\vec{ML}$ with endpoints $M(3, 2)$ and $L(2, 3)$.

2. Find the length of each vector.

a. $\vec{u} = (3, 1)$

b. $\vec{v} = (0, 3)$

c. $\vec{AB}$ with endpoints $A(2, 1)$ and $B(5, 4)$

Answers

1. $\vec{ML} = (-1, 1)$ 2. a. $\sqrt{10}$ b. 3 c. $3\sqrt{2}$

5. Equal Vectors

Definition

equal vectors

Two vectors are **equal** if and only if their corresponding scalar components are equal.

In other words, the vectors $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ are equal if and only if $u_1 = v_1$ and $u_2 = v_2$.

EXAMPLE

18

$\vec{u}$ and $\vec{v}$ are equal vectors with $\vec{u} = (a + b, 3)$, $\vec{v} = (-1, b - a)$. Find a and b .

Solution

$\vec{u} = \vec{v}$, so $a + b = -1$ and $3 = b - a$.

$$\left. \begin{aligned} a + b &= -1 \\ b - a &= 3 \end{aligned} \right\}, \text{ so } b = 1 \text{ and } a = -2.$$

EXAMPLE

19

The vector $\vec{v} = (3, 7)$ has initial point $(2, 4)$. What is its terminal point?

Solution

Let the terminal point of $\vec{v}$ be (x, y) , so

$$(x - 2, y - 4) = (3, 7)$$

$$x - 2 = 3 \text{ and } y - 4 = 7 \text{ by the equality of vectors.}$$

$$\text{So } x = 5 \text{ and } y = 11.$$

EXAMPLE**20**

The initial point and terminal point of $\overrightarrow{MN} = (2, 3)$ are $M(2, k)$ and $N(p, 1)$ respectively. Find $p + k$.

Solution

$$\begin{aligned}\overrightarrow{MN} &= (p - 2, 1 - k) = (2, 3) \\ p - 2 &= 2 \quad \text{and} \quad 1 - k = 3 \\ p &= 4 \quad \text{and} \quad k = -2 \\ \text{So } p + k &= 4 - 2 = 2.\end{aligned}$$

Check Yourself 7

1. $A(2, 1)$, $B(1, -3)$, $C(3, 2)$, and $D(a, b)$ are given. If $\overrightarrow{AC} = \overrightarrow{BD}$, find $|\overrightarrow{AD}|$.
2. Let $\vec{u} = (a, a + 1)$ and $\vec{v} = (3, b)$ such that $\vec{u} = \vec{v}$. Find a and b .

Answers

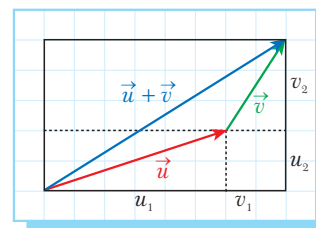
1. 3
2. $a = 3, b = 4$

B. VECTOR OPERATIONS**1. Addition of Vectors**

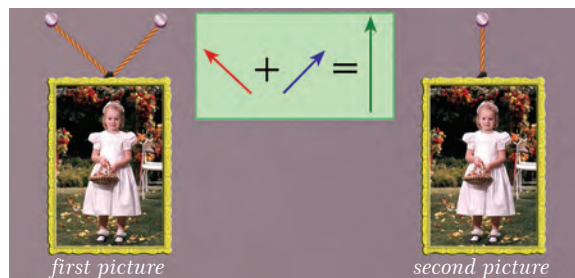
If $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$, then

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2).$$

The figure opposite shows how the analytic definition of addition corresponds to the geometric one.



Two identical pictures are hung on a wall by pieces of string as shown in the figure. Each piece of string exerts a force upon the picture to support its weight. The sum of the forces of the first picture is equal to the force of the second picture. We can represent the forces as vectors. The box shows that the sum of the two vectors of the first picture equals the vector of the second picture.

**EXAMPLE****21**

$\vec{u} = (1, -2)$ and $\vec{v} = (-2, 1)$. Find $\vec{u} + \vec{v}$.

Solution

$$\vec{u} + \vec{v} = (1 - 2, -2 + 1) = (-1, -1).$$

EXAMPLE**22**

$A(-3, 2)$ and $\overrightarrow{AB} = (5, -2)$, are given. Find the coordinates of B .

Solution

$$\begin{aligned}\text{Let us write } B(a, b). \text{ Now, } \overrightarrow{AB} &= (a - (-3), b - 2) = (a + 3, b - 2) \\ (a + 3, b - 2) &= (5, -2).\end{aligned}$$

$a + 3 = 5$ and $b - 2 = -2$ by the equality of vectors, so $a = 2$ and $b = 0$.

Therefore, the coordinates of B are $(2, 0)$.

a. Properties of Vector Addition

Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$, and $\vec{w} = (w_1, w_2)$ be vectors in a plane. Then the following properties hold.

1. The sum of any two vectors in a plane is a vector. (closure property)
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (commutative property)
3. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$ (associative property)
4. $\vec{0}$ is the identity element:

$$\vec{u} + \vec{0} = (u_1 + 0, u_2 + 0) = (u_1, u_2) = \vec{u}.$$
5. $-\vec{u}$ is additive inverse of $\vec{u}$:

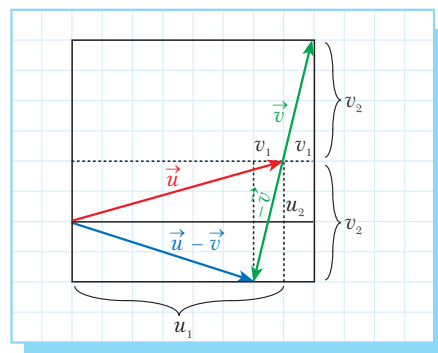
$$\vec{u} + (-\vec{u}) = (u_1 - u_1, u_2 - u_2).$$

2. Subtraction of Vectors

If $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ then

$$\vec{u} - \vec{v} = (u_1 - v_1, u_2 - v_2).$$

The figure opposite shows how the analytic definition of vector subtraction corresponds to the geometric one.



EXAMPLE 23 Subtract $\vec{v} = (3, 2)$ from $\vec{u} = (2, 4)$.

Solution $\vec{u} - \vec{v} = (2 - 3, 4 - 2) = (-1, 2)$

a. Properties of Vector Subtraction

Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$, and $\vec{w} = (w_1, w_2)$ be vectors in a plane. Then the following properties hold.

1. The difference of any two vectors in a plane is a vector.
2. $\vec{u} - \vec{v} \neq \vec{v} - \vec{u}$. Therefore, vector subtraction is not commutative.
3. $\vec{u} - (\vec{v} - \vec{w}) \neq (\vec{u} - \vec{v}) - \vec{w}$. Therefore, vector subtraction is not associative.
4. $\vec{u} - \vec{0} \neq \vec{0} - \vec{u}$. Therefore there is no identity element for subtraction.

EXAMPLE 24 Given $\vec{w} = (1, -3)$, $K(3, 2)$ and $P(-1, 4)$, find $\vec{w} + \vec{KP}$ and $\vec{w} - \vec{KP}$.

Solution

$$\begin{aligned}\overrightarrow{KP} &= (-1 - 3, 4 - 2) = (-4, 2) \\ \vec{w} + \overrightarrow{KP} &= (1, -3) + (-4, 2) = (1 - 4, -3 + 2) = (-3, -1) \\ \vec{w} - \overrightarrow{KP} &= (1, -3) - (-4, 2) = (1 + 4, -3 - 2) = (5, -5)\end{aligned}$$

EXAMPLE

25

$M(1, 4)$, $N(3, 2m)$, $K(m, -1)$, and $|\overrightarrow{MN} + \overrightarrow{NK}| = 13$ are given. Find m .

Solution

$$\begin{aligned}\overrightarrow{MN} + \overrightarrow{NK} &= \overrightarrow{MK} \\ |\overrightarrow{MN} + \overrightarrow{NK}| &= |\overrightarrow{MK}| = 13 \\ \sqrt{(m-1)^2 + (-1-4)^2} &= 13 \\ (m-1)^2 + 25 &= 169 \\ (m-1)^2 &= 144 \\ m-1 &= 12 \text{ or } m-1 = -12 \\ m &= 13 \text{ or } m = -11\end{aligned}$$

3. Multiplication of a Vector by a Scalar

Let $\vec{v} = (v_1, v_2)$ and $c \in \mathbb{R}$, then $c \cdot \vec{v} = (c \cdot v_1, c \cdot v_2)$.

EXAMPLE

26

$\vec{u} = (-3, 2)$ and $\vec{v} = (2, -1)$ are given. Find $-2 \cdot \vec{u}$, $3 \cdot \vec{v}$, and $3 \cdot \vec{u} + 2 \cdot \vec{v}$.

Solution

$$\begin{aligned}-2 \cdot \vec{u} &= (-2 \cdot (-3), (-2) \cdot 2) = (6, -4) \\ 3 \cdot \vec{v} &= (3 \cdot 2, 3 \cdot (-1)) = (6, -3) \\ 3 \cdot \vec{u} + 2 \cdot \vec{v} &= 3 \cdot (-3, 2) + 2 \cdot (2, -1) = (-9, 6) + (4, -2) = (-5, 4)\end{aligned}$$

a. Properties of the Multiplication of a Vector by a Scalar

Let $\vec{u} = (u_1, u_2)$, $\vec{v} = (v_1, v_2)$ and $c, d \in \mathbb{R}$. Then the following properties hold:

1. $c \cdot (\vec{u} + \vec{v}) = c \cdot \vec{u} + c \cdot \vec{v}$. Look at the proof:

$$\begin{aligned}c \cdot (\vec{u} + \vec{v}) &= c \cdot (u_1 + v_1, u_2 + v_2) \\ &= (c \cdot u_1 + c \cdot v_1, c \cdot u_2 + c \cdot v_2) \\ &= c \cdot (u_1, u_2) + c \cdot (v_1, v_2) \\ &= c \cdot \vec{u} + c \cdot \vec{v}.\end{aligned}$$

2. $(c + d) \cdot \vec{u} = c \cdot \vec{u} + d \cdot \vec{u}$

3. $(c \cdot d) \cdot \vec{u} = c \cdot (d \cdot \vec{u}) = d \cdot (c \cdot \vec{u})$

4. $1 \cdot \vec{u} = \vec{u}$

5. $0 \cdot \vec{u} = \vec{0}$

6. $c \cdot \vec{0} = \vec{0}$

7. $|c \cdot \vec{u}| = |c| \cdot |\vec{u}|$.

EXAMPLE

27

Find $|2\vec{u} - 3\vec{v}|$ given $\vec{u} = (-2, 3)$ and $\vec{v} = (1, -1)$.

Solution

$$2\vec{u} - 3\vec{v} = 2(-2, 3) - 3(1, -1) = (-4, 6) - (3, -3) = (-4 - 3, 6 + 3) = (-7, 9)$$

$$\text{Therefore, } |2\vec{u} - 3\vec{v}| = \sqrt{(-7)^2 + 9^2} = \sqrt{49 + 81} = \sqrt{130}.$$

Check Yourself 8

- Find $|\vec{w}|$ given $\vec{u} = (3, -2)$, $\vec{v} = (-1, 4)$, $\vec{w} = (a, b)$ and $\vec{v} - \vec{w} = 3\vec{u}$.
- Find $\vec{x}$ given $2\vec{x} + \vec{y} = (1, 2)$, and $\vec{x} - \vec{y} = (-4, 4)$.

Answers

- $|\vec{w}| = 10\sqrt{2}$
- $\vec{x} = (-1, 2)$

EXAMPLE

28

Find each vector, given $\vec{u} = (-3, 1)$ and $\vec{v} = (2, -2)$.

a. $\vec{u} - 3\vec{v}$

b. $3\vec{u} + 2\vec{v}$

c. $4\vec{u} + \vec{v}$

Solution

a. $\vec{u} - 3\vec{v} = (-3, 1) - 3(2, -2) = (-3, 1) + (-6, 6) = (-3 - 6, 1 + 6) = (-9, 7)$

b. $3\vec{u} + 2\vec{v} = 3(-3, 1) + 2(2, -2) = (-9, 3) + (4, -4) = (-9 + 4, 3 - 4) = (-5, -1)$

c. $4\vec{u} + \vec{v} = 4(-3, 1) + (2, -2) = (-12, 4) + (2, -2) = (-12 + 2, 4 - 2) = (-10, 2)$

4. Standard Base Vectors

Definition

unit vector

A vector of length 1 is called a **unit vector**.

For example, the vector $\vec{w} = (\frac{3}{5}, \frac{4}{5})$ is a unit vector.

For any non-zero vector $\vec{u} = (u_1, u_2)$, $\frac{\vec{u}}{|\vec{u}|}$ is a unit vector because $\frac{1}{|\vec{u}|} = \frac{1}{\sqrt{u_1^2 + u_2^2}}$ and

$$\frac{\vec{u}}{|\vec{u}|} = \frac{1}{|\vec{u}|} \cdot \vec{u} = \left(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}} \right).$$

$$\text{So } \left| \frac{\vec{u}}{|\vec{u}|} \right| = \sqrt{\left(\frac{u_1}{\sqrt{u_1^2 + u_2^2}} \right)^2 + \left(\frac{u_2}{\sqrt{u_1^2 + u_2^2}} \right)^2} = \sqrt{\frac{u_1^2}{u_1^2 + u_2^2} + \frac{u_2^2}{u_1^2 + u_2^2}} = \sqrt{\frac{u_1^2 + u_2^2}{u_1^2 + u_2^2}} = 1.$$

We sometimes use $\frac{\vec{u}}{|\vec{u}|}$ to find the direction of $\vec{u}$.

There are two important unit vectors, $\vec{i}$ and $\vec{j}$, defined as $\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$.

These vectors are special because we can use them to express any vector.

We call these vectors **standard base vectors**.

For example, let us express the vector $\vec{v} = (v_1, v_2)$ in terms of $\vec{i}$ and $\vec{j}$:
 $\vec{v} = (v_1, v_2) = v_1 \cdot \vec{i} + v_2 \cdot \vec{j}$.

In this expression,

$v_1 \cdot \vec{i}$ is the horizontal component,

$v_2 \cdot \vec{j}$ is the vertical component,

v_1 is the horizontal scalar component, and

v_2 is the vertical scalar component.

We can prove the proposition above by using algebraic operations on vectors and the properties of real numbers:

$$\begin{aligned}\vec{v} = (v_1, v_2) &= (1 \cdot v_1 + 0, 0 + 1 \cdot v_2) = (1 \cdot v_1, 0) + (0, 1 \cdot v_2) \\ &= v_1 \cdot (1, 0) + v_2 \cdot (0, 1) \\ &= v_1 \cdot \vec{i} + v_2 \cdot \vec{j}.\end{aligned}$$

EXAMPLE

29

- a. Write the vector $\vec{u} = (5, -8)$ in terms of $\vec{i}$ and $\vec{j}$.
 b. If $\vec{u} = 3 \cdot \vec{i} + 2 \cdot \vec{j}$ and $\vec{v} = -\vec{i} + 6 \cdot \vec{j}$, find $2 \cdot \vec{u} + 3 \cdot \vec{v}$.

Solution

a. $\vec{u} = 5 \cdot \vec{i} + (-8) \cdot \vec{j} = 5\vec{i} - 8\vec{j}$.

- b. The properties of addition and scalar multiplication of vectors show that we can manipulate vectors in the same way we manipulate algebraic expressions. Therefore,

$$\begin{aligned}2 \cdot \vec{u} + 3 \cdot \vec{v} &= 2 \cdot (3 \cdot \vec{i} + 2 \cdot \vec{j}) + 3 \cdot (-\vec{i} + 6 \cdot \vec{j}) \\ &= (6 \cdot \vec{i} + 4 \cdot \vec{j}) + (-3 \cdot \vec{i} + 18 \cdot \vec{j}) \\ &= (6 - 3) \cdot \vec{i} + (4 + 18) \cdot \vec{j} \\ &= 3 \cdot \vec{i} + 22 \cdot \vec{j}.\end{aligned}$$

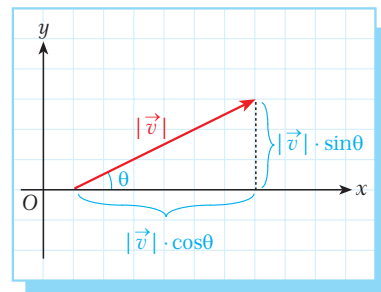


$$\begin{aligned}\cos 60^\circ &= \sin 30^\circ = \frac{1}{2} \\ \sin 60^\circ &= \cos 30^\circ = \frac{\sqrt{3}}{2} \\ \sin 45^\circ &= \cos 45^\circ = \frac{\sqrt{2}}{2} \\ \sin 90^\circ &= \cos 0^\circ = 1 \\ \sin 0^\circ &= \cos 90^\circ = 0 \\ \cos (180 - \alpha) &= -\cos \alpha \\ \sin (180 - \alpha) &= \sin \alpha \\ \cos (90 - \alpha) &= \sin \alpha \\ \sin (90 - \alpha) &= \cos \alpha\end{aligned}$$

Let $\vec{v}$ be a vector in the plane with its initial point at the origin. Let θ be the positive angle between the positive x -axis and $\vec{v}$ (see the figure). If we know the length and direction of $\vec{v}$, then we can resolve the vector into horizontal and vertical components in terms of θ :

$\vec{v}$ has length $|\vec{v}|$, and $\vec{v} = (v_1, v_2) = v_1 \cdot \vec{i} + v_2 \cdot \vec{j}$.

So $v_1 = |\vec{v}| \cdot \cos \theta$ and $v_2 = |\vec{v}| \cdot \sin \theta$.

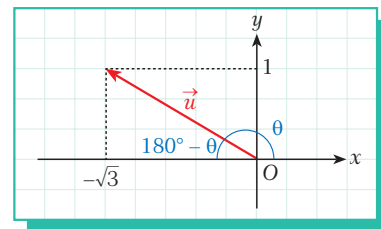


EXAMPLE

30

- a. $|\vec{v}| = 4$ cm and the angle between $\vec{v}$ and the positive x -axis is 60° . Find the horizontal and vertical components of $\vec{v}$ and express $\vec{v}$ in terms of $\vec{i}$ and $\vec{j}$.
 b. Find the angle θ between the vector $\vec{u} = -\sqrt{3} \cdot \vec{i} + \vec{j}$ and the positive x -axis.

- Solution** a. We have $\vec{v} = (v_1, v_2)$, where the scalar components are given by $v_1 = 4 \cdot \cos 60^\circ = 2$ and $v_2 = 4 \cdot \sin 60^\circ = 2\sqrt{3}$. Therefore, the horizontal component is $2 \cdot \vec{i}$ and the vertical component is $2\sqrt{3} \cdot \vec{j}$. Therefore, $\vec{v} = 2 \cdot \vec{i} + 2\sqrt{3} \cdot \vec{j}$.
- b. From the figure we see that θ has the property that
- $$\tan(180^\circ - \theta) = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$
- Thus $180 - \theta = 30^\circ$, and so $\theta = 150^\circ$.



Check Yourself 9

- Express $\vec{v} = (3, 2)$ in terms of standard base vectors.
- Resolve the vector $\vec{u}$ with length 3 cm into its components if the angle between the positive x -axis and $\vec{u}$ is 120° .
- Find the angle θ between the vector $\vec{v} = -3 \cdot \vec{i} - 3\sqrt{3} \cdot \vec{j}$ and the positive x -axis.

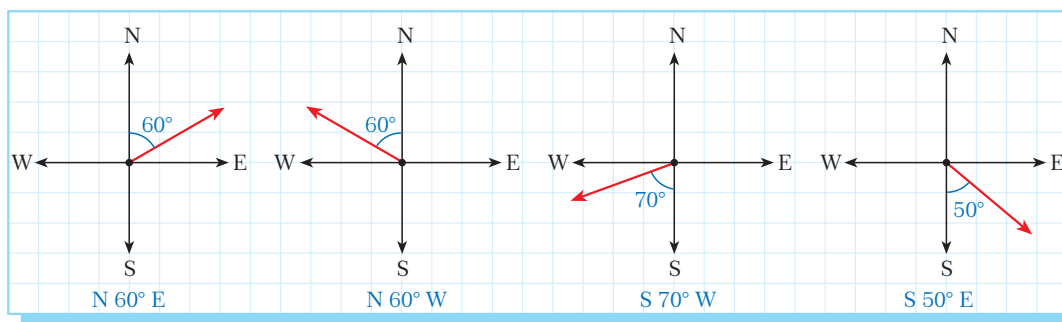
Answers

- $\vec{v} = 3 \cdot \vec{i} + 2 \cdot \vec{j}$
- $\vec{u}_x = (-\frac{3}{2}, 0)$, $\vec{u}_y = (0, \frac{3\sqrt{3}}{2})$
- $\theta = 240^\circ$

C. VECTOR APPLICATIONS

As we have seen, vectors describe quantities that have both magnitude and direction. They have a wide range of applications, such as in navigation, mechanics, and engineering.

In navigation, the direction of movement of an object such as a ship is usually given as a **bearing**, that is, as an acute angle measured from due north or due south. For example, the bearing N 60° E stands for a direction that points 60° to the east of due north. Look at some more examples of bearings in the figure:

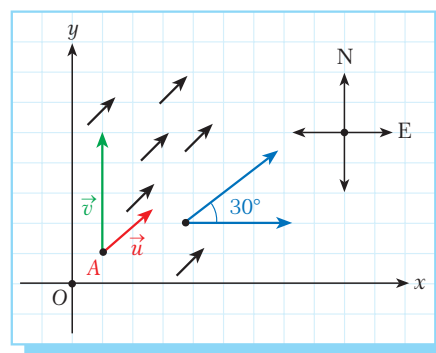


Note that N 60° E is not the same as E 60° N.

The speed of a moving object along a bearing is called the **velocity** of the object. We can see that velocity has direction (a bearing) and magnitude (speed). Therefore we can use a vector to represent a velocity.

In the figure, the vector $\vec{u}$ represents the velocity of wind blowing in the direction N 60° E, and the vector $\vec{v}$ represents the velocity of an airplane flying through this wind at point A. It is obvious that the wind affects both the speed and the direction of the airplane.

From the figure we can see that actual velocity of the plane (relative to the ground) is given by the vector $\vec{w} = \vec{u} + \vec{v}$.



EXAMPLE

31

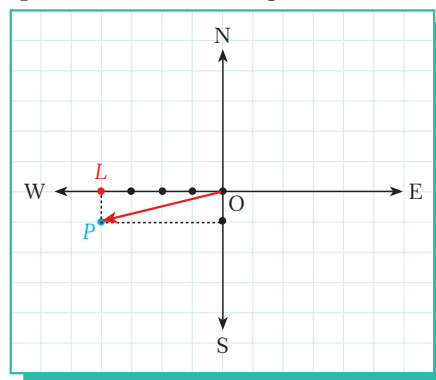
An airplane is headed due west at an air speed of 400 km/h and the wind is blowing from the north at 100 km/h. Find the resultant velocity of the airplane relative to the ground.

Solution

We first construct a rectangular coordinate system to indicate the four directions east, west, north and south.

Look at the figure. We can draw a vector $\vec{LP}$ to represent the direction and magnitude of the air velocity of the plane, and a vector $\vec{OP}$ to represent the direction and magnitude of the wind velocity. We can now expect the resultant velocity to be in the direction of $\vec{OP}$ with a magnitude proportional to the length of $\vec{OP}$. Therefore, we have $|\vec{OP}| = \sqrt{400^2 + 100^2} \cong 412$.

So the velocity of the airplane is approximately 412 km/h along $\vec{OP}$.



EXAMPLE

32

A woman wants to move by boat from one shore of a straight river to the point directly opposite on the other shore. The speed of the boat in still water is 12 km/h and the river is flowing east at the rate of 6 km/h. In what direction should the woman head the boat in order to arrive at the point directly opposite?

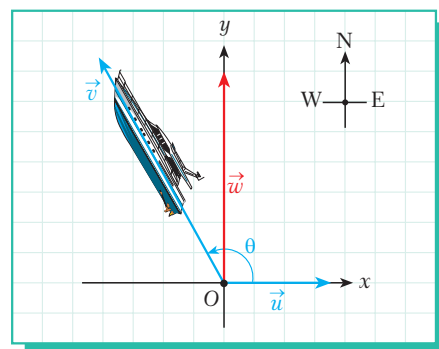
Solution

Let us choose a coordinate system with the origin at the initial position of the boat as shown in the figure.

Let $\vec{u}$ and $\vec{v}$ represent the velocities of the river and the boat, respectively.

$$\begin{aligned} \text{Then } \vec{u} &= 6 \cdot \vec{i} \text{ and } |\vec{v}| = 12, \\ \vec{v} &= (12 \cdot \cos \theta) \cdot \vec{i} + (12 \cdot \sin \theta) \cdot \vec{j} \end{aligned}$$

where the angle θ is as shown in the figure.



The actual velocity of the boat is now

$$\begin{aligned}\vec{w} &= \vec{u} + \vec{v} = 6\vec{i} + (12\cos\theta)\vec{i} + (12\sin\theta)\vec{j} \\ &= (6 + 12\cos\theta)\vec{i} + (12\sin\theta)\vec{j}.\end{aligned}$$

Since the woman wants to land at a point directly opposite, her direction should have horizontal component 0.

$$\begin{aligned}6 + 12\cos\theta &= 0 \\ \cos\theta &= -\frac{1}{2} \\ \theta &= 120^\circ\end{aligned}$$

Therefore, the woman should head the boat in the direction N 30° W.

Check Yourself 10

1. A jet plane is flying in the direction N 20° E with a speed of 500 km/h. Find the east and north components of the velocity.
2. A vector is 4 cm long and makes an angle of 60° with the positive x -axis. Resolve the vector into its components.

Answers

1. $500 \cdot \cos 70^\circ \cdot \vec{i}$, $500 \cdot \sin 70^\circ \cdot \vec{j}$ 2. $\vec{u} = 2\vec{i} + 2\sqrt{3}\vec{j}$

D. PARALLEL VECTORS

We know from the geometrical analysis of vectors that two non-zero vectors are parallel if and only if multiplying one of them by a suitable scalar equals the other, that is,

for any $c \neq 0$, $\vec{u} \neq 0$, and $\vec{v} \neq 0$, $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} = c \cdot \vec{v}$.

It follows that if $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$, then $(u_1, u_2) = (c \cdot v_1, c \cdot v_2)$.

So $\vec{u} \parallel \vec{v}$ if and only if $\frac{u_1}{v_1} = \frac{u_2}{v_2} = c$.

EXAMPLE 33 Show that $\vec{u} = (2, 1)$ and $\vec{v} = (1, \frac{1}{2})$ are parallel.

Solution $\frac{2}{1} = \frac{1}{\frac{1}{2}} = 2$, so they are parallel.

EXAMPLE 34 Find the relation between x and y given $A(3, -1)$, $B(2, 3)$, $C(5, -4)$, $D(x, y)$, and $\vec{CD} \parallel \vec{AB}$.

Solution $\vec{AB} = (2 - 3, 3 - (-1)) = (-1, 4)$
 $\vec{CD} = (x - 5, y + 4)$
 $\vec{CD} \parallel \vec{AB}$ so $\frac{x-5}{-1} = \frac{y+4}{4}$.
 $4x - 20 = -y - 4$, so $4x + y - 16 = 0$.

EXAMPLE 35 A triangle ABC has vertices $A(3, 2)$, $B(-3, -2)$, and $C(2, -3)$. E and F are the midpoints of sides AB and AC respectively. Find the coordinates of E and F .

Solution Let us write $E(x, y)$ and $F(m, n)$.

$$\begin{array}{ll} \vec{BE} = \vec{EA} & \text{and} \quad \vec{CF} = \vec{FA} \\ \vec{BE} = (x + 3, y + 2) & \vec{CF} = (m - 2, n + 3) \\ \vec{EA} = (3 - x, 2 - y) & \vec{FA} = (3 - m, 2 - n) \\ x + 3 = 3 - x & m - 2 = 3 - m \\ 2x = 0 & 2m = 5 \\ x = 0 & m = \frac{5}{2} \\ y + 2 = 2 - y & n + 3 = 2 - n \\ 2y = 0 & 2n = -1 \\ y = 0 & n = -\frac{1}{2} \end{array}$$

Remember!
If two parallel vectors $\vec{a}$ and $\vec{b}$ have at least one point in common, then $\vec{a}$ and $\vec{b}$ lie on the same straight line (they are collinear).
For instance, if $\vec{AB} = k \cdot \vec{BC}$ for some $k \in \mathbb{R} \setminus \{0\}$, then A , B , and C are collinear.

Therefore the coordinates are $E(0, 0)$ and $F(\frac{5}{2}, -\frac{1}{2})$.

Check Yourself 11

1. Show that $\vec{u} = (-2, -3)$ and $\vec{v} = (4, 6)$ are parallel.
2. Find the value of k if $\vec{u} = (1, k)$ and $\vec{v} = (3, 6)$ are parallel.

Answers

1. check the scalar components
2. $k = 2$

E. LINEAR COMBINATION OF VECTORS

1. Linear Combination of Vectors

Definition

linear combination of vectors

Let $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k$ be vectors in the plane and let $c_1, c_2, \dots, c_k$ be scalars.

The expression $c_1 \cdot \vec{u}_1 + c_2 \cdot \vec{u}_2 + \dots + c_k \cdot \vec{u}_k$ is called a **linear combination** of the vectors.

For example, $w = 2 \cdot \vec{u} - 3 \cdot \vec{v}$ is a linear combination of vectors.

EXAMPLE**36**Find the vector $\vec{w}$ if $\vec{w} = 2\vec{u} - 3\vec{v}$, $\vec{u} = (5, -2)$, and $\vec{v} = (1, 3)$.**Solution**

$$\vec{w} = 2\vec{u} - 3\vec{v} = 2 \cdot (5, -2) - 3 \cdot (1, 3) = (10, -4) + (-3, -9) = (7, -13)$$

EXAMPLE**37**Express $\vec{v} = (12, 5)$ as a linear combination of the vectors $\vec{u}_1 = (2, 1)$ and $\vec{u}_2 = (3, 2)$.**Solution**Let $c_1, c_2 \in \mathbb{R}$. Then $\vec{v} = c_1\vec{u}_1 + c_2\vec{u}_2$.

$$(12, 5) = c_1 \cdot (2, 1) + c_2 \cdot (3, 2)$$

$$(12, 5) = (2 \cdot c_1, c_1) + (3 \cdot c_2, 2 \cdot c_2)$$

$$(12, 5) = (2 \cdot c_1 + 3 \cdot c_2, c_1 + 2 \cdot c_2)$$

$$12 = 2 \cdot c_1 + 3 \cdot c_2 \quad \text{and} \quad 5 = c_1 + 2 \cdot c_2$$

$$12 = 2 \cdot (5 - 2 \cdot c_2) + 3 \cdot c_2 \quad 5 - 2 \cdot c_2 = c_1$$

$$12 = 10 - 4 \cdot c_2 + 3 \cdot c_2$$

$$2 = -c_2 \quad (1)$$

$$c_2 = -2 \quad (2)$$

$$c_1 = 5 + 4 = 9$$

Using (1) and (2) gives $\vec{v} = 9 \cdot \vec{u}_1 - 2 \cdot \vec{u}_2$.

If each element of a set of vectors V can be expressed as a linear combination of vectors $\vec{u}$ and $\vec{v}$, then we say that $\vec{u}$ and $\vec{v}$ **span** the set V .

Note

We cannot express any vector as a linear combination of two parallel vectors.

EXAMPLE**38**Express the vector $\vec{x} = (19, 13)$ as a linear combination of $\vec{u} = (2, -3)$, $\vec{v} = (1, 2)$ and $\vec{w} = (5, 4)$.**Solution**Let c_1, c_2 , and $c_3 \in \mathbb{R}$. Then $\vec{x} = c_1\vec{u} + c_2\vec{v} + c_3\vec{w}$.

$$(19, 13) = c_1 \cdot (2, -3) + c_2 \cdot (1, 2) + c_3 \cdot (5, 4)$$

$$(19, 13) = (2 \cdot c_1, -3 \cdot c_1) + (c_2, 2 \cdot c_2) + (5 \cdot c_3, 4 \cdot c_3)$$

$$(19, 13) = (2c_1 + c_2 + 5c_3, -3c_1 + 2c_2 + 4c_3)$$

$$19 = 2c_1 + c_2 + 5c_3 \quad \text{and} \quad 13 = -3c_1 + 2c_2 + 4c_3$$

$$c_2 = 19 - 2c_1 - 5c_3 \quad \text{and} \quad 13 = -3c_1 + 2 \cdot 19 - 2 \cdot 2c_1 - 2 \cdot 5c_3 + 4c_3$$

$$13 = -3c_1 - 4c_1 - 6c_3 + 38$$

$$7c_1 + 6c_3 = 25$$

There are many solutions to this equation.

Let us choose $c_1 = 1$, then $c_2 = 2$ and $c_3 = 3$, so $\vec{x} = \vec{u} + 2\vec{v} + 3\vec{w}$. This is one solution to the problem.

2. Linearly Dependent and Independent Vectors (*OPTIONAL*)

Definition

linearly dependent and independent vectors

A set of vectors $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ in a vector space V is called **linearly independent** if the vector equation $c_1 \cdot \vec{v}_1 + c_2 \cdot \vec{v}_2 + \dots + c_k \cdot \vec{v}_k = \vec{0}$ has only the trivial solution $c_1 = 0, c_2 = 0, \dots, c_k = 0$.

If any of c_i is different from zero, then the set S is called **linearly dependent**.

EXAMPLE

39

Show that $S = \{\vec{v}, \vec{w}\}$, is linearly independent if $\vec{v} = (1, 0)$, and $\vec{w} = (0, 1)$.

Solution

$$\begin{aligned} c_1 \cdot \vec{v} + c_2 \cdot \vec{w} &= \vec{0} \\ c_1 \cdot (1, 0) + c_2 \cdot (0, 1) &= (0, 0) \\ (c_1, 0) + (0, c_2) &= (0, 0) \\ (c_1, c_2) &= (0, 0) \\ c_1 = 0 \text{ and } c_2 &= 0 \end{aligned}$$

So S is linearly independent by the definition.

EXAMPLE

40

Show that $S = \{\vec{u}, \vec{v}, \vec{w}\}$, is linearly dependent if $\vec{u} = (1, 0)$, $\vec{v} = (0, 1)$ and $\vec{w} = (-2, 5)$.

Solution

Let $c_1, c_2, c_3 \in \mathbb{R}$ and $c_1 \cdot \vec{u} + c_2 \cdot \vec{v} + c_3 \cdot \vec{w} = \vec{0}$.

$$\begin{aligned} c_1 \cdot (1, 0) + c_2 \cdot (0, 1) + c_3 \cdot (-2, 5) &= (0, 0) \\ (c_1, 0) + (0, c_2) + (-2c_3, 5c_3) &= (0, 0) \\ (c_1 - 2c_3, c_2 + 5c_3) &= (0, 0) \\ c_1 - 2c_3 = 0 &\quad \text{and} \quad c_2 + 5c_3 = 0 \\ c_1 = 2c_3 &\quad \text{and} \quad c_2 = -5c_3 \end{aligned}$$

There are infinitely many solutions to this set of equations. For example, if

$c_3 = 1$, then $c_1 = 2$ and $c_2 = -5$.

These values are non-zero, so the vector set is linearly dependent.

Note

In a plane, two non-parallel and non-zero vectors are linearly independent but two parallel vectors are linearly dependent.

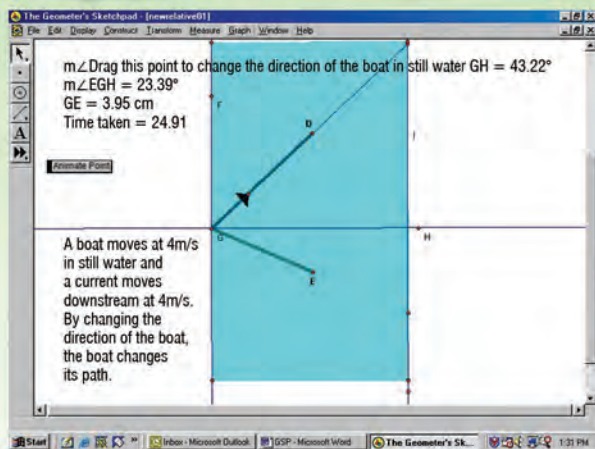
Check Yourself 12

- Express $\vec{u} = (0, 1)$ as a linear combination of $\vec{v} = (1, 1)$ and $\vec{w} = (-1, 2)$.
- Show that $\vec{a} = (2, 0)$ and $\vec{b} = (0, 2)$ are linearly independent.

Answers

- $\vec{u} = \frac{1}{3} \cdot \vec{v} + \frac{1}{3} \cdot \vec{w}$
- solve the equation $\vec{u} = x \cdot \vec{v} + y \cdot \vec{w}$

Computer Applications



The screen opposite shows a simple problem illustrated using *The Geometer's Sketchpad*. Suppose a current flows at a certain velocity w downstream. A boat moves at a constant speed v . Which direction the boat take in order to reach the other side of the river in the shortest possible time?

The Geometer's Sketchpad calculates the time as we move the point H on the screen to set the direction of the boat. Using the *Sketchpad* we can move H to find the shortest possible time in the problem.

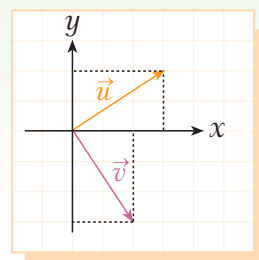
We can use computer applications such as *The Geometer's Sketchpad*, *Cabri Geometry*, or *Javascript sketchpad* to sketch vectors and solve problems. We can use an application to a problem, and then change certain values to see their effect. We can also use a computer application to add and subtract vectors, and to multiply a vector by a scalar.

Project:

Use *The Geometer's Sketchpad*, *Cabri Geometry*, or *Javascript sketchpad* to sketch the vectors

$$2\vec{u}, -\vec{v}, \vec{u} + \vec{v}, \vec{u} - 3\vec{v} \text{ and}$$

$$\overrightarrow{PQ}, \overrightarrow{QP} \text{ if } P(3, -4) \text{ and } Q(4, -3).$$



EXERCISES 2

A. Basic Concepts of Vectors in the Analytic Plane

- Plot the points $A(-1, 1)$, $B(2, -1)$, $C(3, 1)$, and $D(-3, -1)$ in the plane.

- Sketch the position vector of the vector with the given endpoints.

a. $\overrightarrow{AB}$

b. $\overrightarrow{CD}$

c. $\overrightarrow{EF}$

$A(-2, -3)$

$C(1, -5)$

$E(0, -3)$

$B(4, -1)$

$D(0, 2)$

$F(-4, 2)$

- Find the length of $\overrightarrow{MN}$ given $M(1, 4)$ and $N(-2, -1)$.

- The figure shows the vectors $\vec{u}$ and $\vec{v}$. Sketch the following vectors.

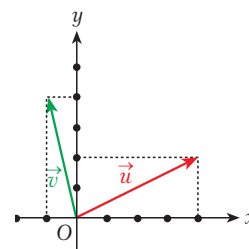
a. $2\vec{v}$

b. $-\vec{u}$

c. $\vec{u} + \vec{v}$

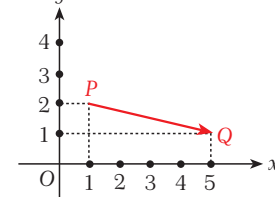
d. $\vec{u} - 2\vec{v}$

e. $2\vec{u} + \vec{v}$



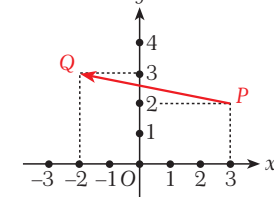
- Describe the vector with initial point P and terminal point Q .

a.



c. $P(3, 2)$, $Q(8, 9)$

b.



d. $P(-1, 3)$, $Q(1, 0)$

6. Find the vectors $\vec{a}$ and $\vec{b}$ if $2\vec{a} - 3\vec{b} = (4, 2)$ and $\vec{a} + \vec{b} = (2, 1)$.

B. Vector Operations

7. Find $\vec{u} + \vec{v}$, $\vec{u} - \vec{v}$, $2\vec{u}$, $2\vec{u} - \vec{v}$, $\vec{u} + 3\vec{v}$, and $-4\vec{u} + 3\vec{v}$ for the given vectors $\vec{u}$ and $\vec{v}$.

- a. $\vec{u} = (2, 6)$, $\vec{v} = (1, 3)$
b. $\vec{u} = (-2, 3)$, $\vec{v} = (8, -2)$
c. $\vec{u} = (1, 0)$, $\vec{v} = (0, -2)$
d. $\vec{u} = \vec{i}$, $\vec{v} = \vec{j}$
e. $\vec{u} = -3\vec{i} + \vec{j}$, $\vec{v} = \vec{i} - \vec{j}$
f. $\vec{u} = 7\vec{i} + 5\vec{j}$, $\vec{v} = \vec{j} - \vec{i}$

8. Find $|\vec{u}|$, $|\vec{v}|$, $|2\vec{v}|$, $|\frac{1}{3}\vec{v}|$, $|\vec{u} + \vec{v}|$, and $|\vec{u} - \vec{v}|$ for the given vectors $\vec{u}$ and $\vec{v}$.

- a. $\vec{u} = 3\vec{i} + \vec{j}$, $\vec{v} = -\vec{i} + 2\vec{j}$
b. $\vec{u} = 2\vec{i} - \vec{j}$, $\vec{v} = -\vec{i} - \vec{j}$
c. $\vec{u} = (2, 3)$, $\vec{v} = (0, 1)$
d. $\vec{u} = (3, 4)$, $\vec{v} = (2, 5)$

9. Find the horizontal and vertical scalar components of the vector with the given length and angle with the positive x -axis. Express the vector in terms of standard base vectors.

- a. $|\vec{v}| = 20$, $\theta = 30^\circ$
b. $|\vec{v}| = 30$, $\theta = 120^\circ$
c. $|\vec{v}| = 1$, $\theta = 225^\circ$
d. $|\vec{u}| = 80$, $\theta = 135^\circ$
e. $|\vec{v}| = 4$, $\theta = 10^\circ$
f. $|\vec{u}| = \sqrt{3}$, $\theta = 300^\circ$

10. Given $\vec{AB} = -7\vec{i} + 2\vec{j}$ and $B(3, 11)$, find the coordinates of point A.

11. Given $\vec{AB} = 5\vec{i} + 6\vec{j}$ and $B(-4, 8)$, find the coordinates of point A.

12. $\vec{u} = 3\vec{i} + 4\vec{j}$ and $\vec{v} = 4\vec{i} + \vec{j}$ are given. Which vector is the longest?

13. Given $\vec{u} = 3\vec{i} + 4\vec{j}$, calculate $|\frac{1}{2}\vec{u} - \vec{i}|$.

C. Vector Applications

14. A jogger runs with a constant speed of 6 km/h in the opposite direction to the wind. Find the actual velocity of the jogger if the wind blows at a speed of 3 km/h due west.

15. The jogger in question 14 above heads due north. What is the actual velocity of the jogger?

16. A swimmer can swim with a velocity of 1.4 m/s in still water. The current of a river is 1 m/s due west. The swimmer swims due west in the river. Find the swimmer's actual velocity.

17. The swimmer in question 16 swims due east (against the current). Find the swimmer's actual velocity.

18. A pilot heads his airplane due west. The airplane has a speed of 425 km/h in still air. The wind is blowing due north with a speed of 40 km/h.

- Find the actual velocity of the airplane.
- Find the actual speed of the plane.

19. A boat heads in the direction N 36° E along a river. The speed of the boat in still water is 16 km/h. The river is flowing directly south. It is observed that the actual direction of the boat is directly east. Find the speed of the river current and the actual speed of the boat.

D. Parallel Vectors

20. Show that $\vec{u} = (a, b)$ and $\vec{v} = (2a, 2b)$ are parallel.

21. $\vec{u} \parallel \vec{v}$, $\vec{u} = (1, k - 3)$, and $\vec{v} = (k, k - 4)$ are given. Find the value of k .

22. In a triangle ABC , the vertices are $A(-2, 3)$, $B(0, 1)$ and $C(4, 1)$. Points $D(-1, 2)$ and $E(1, 2)$ are on the sides AB and AC respectively. Show that $\vec{DE} \parallel \vec{BC}$.

23. In a triangle ABC , the vertices are $A(1, 3)$, $B(2, 1)$, and $C(3, 2)$. $E(\frac{3}{2}, 2)$ is on the side AB . Find the coordinates of F if F is on AC and $\vec{EF} \parallel \vec{BC}$.

E. Linear Combination of Vectors

24. $\vec{a} = (13, 2)$, $\vec{b} = (11, 6)$ and $\vec{c} = (1, 1)$ are given. If $\vec{a} + \vec{b} + \vec{u} = 4\vec{c} - \vec{u}$, find $\vec{u}$.

25. Show that the vectors $\vec{a} = (1, 2)$, $\vec{b} = (3, 1)$ and $\vec{c} = (1, 1)$ are linearly dependent.

26. Let $\vec{u}$, $\vec{v}$ and $\vec{w}$ be unit vectors with angles of $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{3\pi}{4}$ respectively with the positive x -axis. Express $\vec{u}$ as a linear combination of $\vec{v}$ and $\vec{w}$.

27. Show that any three non-parallel, non-zero vectors in a plane are linearly dependent.

28. $\vec{a} = (1, -2)$, $\vec{b} = (3, -1)$, and $\vec{c} = (-1, 7)$ are given. Express $\vec{a} + \vec{b} + \vec{c}$ in terms of $\vec{a}$ and $\vec{b}$.



THE DOT PRODUCT OF TWO VECTORS

We have seen how to add and subtract vectors, and how to multiply vectors by a scalar.

In this section we will introduce another operation on vectors, called the **dot product**.

The dot product is sometimes called the scalar product because the resulting product is a number and not a vector. It has applications in mathematics, as well as in engineering and physics.

A. DOT PRODUCT

Definition

dot product (scalar product, or Euclidean inner product)

Let $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ be two vectors in the plane.

The **dot product** of $\vec{u}$ and $\vec{v}$, denoted by $\vec{u} \cdot \vec{v}$, is defined by

$$\vec{u} \cdot \vec{v} = u_1 \cdot v_1 + u_2 \cdot v_2.$$

Thus, to find the dot product of two vectors we multiply the corresponding scalar components and then add them together.

EXAMPLE

41

Find the dot product of the given vectors.

a. $\vec{u} = (-2, 3)$ and $\vec{v} = (5, 4)$

b. $\vec{u} = -\vec{i} + \vec{j}$ and $\vec{v} = 5\vec{i} + \vec{j}$

Solution

a. $\vec{u} \cdot \vec{v} = -2 \cdot 5 + 3 \cdot 4 = -10 + 12 = 2$

b. $\vec{u} \cdot \vec{v} = -1 \cdot 5 + 1 \cdot 1 = -5 + 1 = -4$

1. Properties of the Dot Product

The definition of the dot product gives us the following properties.

1. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$ (commutative property)

2. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ (associative property)

3. $c \cdot (\vec{u} \cdot \vec{v}) = (c \cdot \vec{u}) \cdot \vec{v}$

4. $\vec{u} \cdot \vec{u} = |\vec{u}|^2$

5. $\vec{u} \cdot \vec{v} \geq 0$, and $\vec{u} \cdot \vec{u} = 0$ if and only if $\vec{u} = \vec{0}$.

Proof

1. $\vec{u} \cdot \vec{v} = u_1 \cdot v_1 + u_2 \cdot v_2 = v_1 \cdot u_1 + v_2 \cdot u_2 = \vec{v} \cdot \vec{u}$, by the commutative property of real numbers.

$$\begin{aligned} 2. \vec{u} \cdot (\vec{v} + \vec{w}) &= (u_1, u_2) \cdot (v_1 + w_1, v_2 + w_2) = u_1 \cdot (v_1 + w_1) + u_2 \cdot (v_2 + w_2) \\ &= u_1 \cdot v_1 + u_1 \cdot w_1 + u_2 \cdot v_2 + u_2 \cdot w_2 \\ &= u_1 \cdot v_1 + u_2 \cdot v_2 + u_1 \cdot w_1 + u_2 \cdot w_2 \\ &= \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w} \end{aligned}$$

3. $c \cdot (\vec{u} \cdot \vec{v}) = c \cdot (u_1 \cdot v_1 + u_2 \cdot v_2) = c \cdot u_1 \cdot v_1 + c \cdot u_2 \cdot v_2 = (c \cdot \vec{u}) \cdot \vec{v}$
4. $\vec{u} \cdot \vec{u} = u_1 \cdot u_1 + u_2 \cdot u_2 = u_1^2 + u_2^2 = |\vec{u}|^2$, so $\sqrt{\vec{u} \cdot \vec{u}} = |\vec{u}|$.
5. This proof is left as an exercise for you.

EXAMPLE

42

Find the length of the vector $\vec{u} = (2, -2)$ by using the dot product.

Solution $\vec{u} \cdot \vec{u} = 2 \cdot 2 + (-2) \cdot (-2) = 4 + 4 = 8$. Therefore, $|\vec{u}| = \sqrt{8}$.

Check Yourself 13

1. Find the dot product of $\vec{u} = 3 \cdot \vec{j}$ and $\vec{v} = \vec{i} + \vec{j}$.
2. Find the quantity $(\vec{u} + \vec{v})(\vec{u} - 3 \cdot \vec{v})$ given $\vec{u} = \vec{j}$, $\vec{v} = \vec{i}$.

Answers

1. 3
2. -2

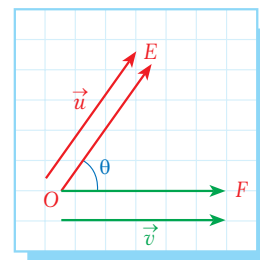
B. ANGLE BETWEEN TWO VECTORS

1. Angle Between Two Vectors

Definition

angle between two vectors

Let $\vec{u} = \overrightarrow{OE}$ and $\vec{v} = \overrightarrow{OF}$ be two non-zero vectors. The angle $\angle EOF$ is called the **angle between $\vec{u}$ and $\vec{v}$** . We use θ to represent the smaller angle between $\vec{u}$ and $\vec{v}$ when their initial points coincide.



Theorem

dot product theorem

Let θ be the angle measure between two non-zero vectors $\vec{u}$ and $\vec{v}$. Then

$$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta.$$

Proof

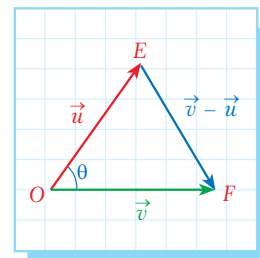
The proof is a nice application of the law of cosines.

As we know, $|\vec{v} - \vec{u}|^2 = (\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) = \vec{v}^2 + \vec{u}^2 - 2 \cdot \vec{v} \cdot \vec{u}$ (1).

Applying the cosine law to triangle EOF in the figure,

$$|\vec{v} - \vec{u}|^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2 \cdot |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta \quad (2) \quad (0 < \theta < \pi)$$

$$(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) = |\vec{u}|^2 + |\vec{v}|^2 - 2 \cdot |\vec{u}| \cdot |\vec{v}| \cos \theta.$$



cosine law:
 $a^2 = b^2 + c^2 - 2bc \cdot \cos \angle A$

By (1) and (2),

$$\vec{v}^2 - 2 \cdot \vec{v} \cdot \vec{u} + \vec{u}^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2 \cdot |\vec{u}| \cdot |\vec{v}| \cos \theta$$

$$2 \cdot \vec{v} \cdot \vec{u} = 2 \cdot |\vec{u}| \cdot |\vec{v}| \cos \theta.$$

$$\text{Therefore, } \vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta.$$

EXAMPLE

43

Given $\vec{u} = (2, -2)$, $\vec{v} = (5, 8)$ and $\vec{w} = (-4, 3)$, find the following.

- a. $\vec{u} \cdot \vec{v}$ b. $(\vec{u} \cdot \vec{v}) \cdot \vec{w}$ c. $\vec{u} \cdot (2 \cdot \vec{v})$ d. $|\vec{w}|^2$

Solution

- a. By the definition of the dot product we have $\vec{u} \cdot \vec{v} = 2 \cdot 5 + (-2) \cdot 8 = -6$.
 b. Using the result from (a) we have $(\vec{u} \cdot \vec{v}) \cdot \vec{w} = -6 \cdot \vec{w} = -6 \cdot (-4, 3) = (24, -18)$.
 c. By property 3 of the dot product we have $\vec{u} \cdot (2 \cdot \vec{v}) = 2 \cdot (\vec{u} \cdot \vec{v}) = 2 \cdot (-6) = -12$.
 d. By property 4 of the dot product we have $|\vec{w}|^2 = \vec{w} \cdot \vec{w} = (-4)^2 + 3^2 = 25$.

One of the most important uses of the dot product is to find the angle between two vectors if the scalar components of the vectors are given. We simply calculate the angle by solving the equation given by the dot product theorem for $\cos \theta$. Let us state this important result clearly.

FINDING THE ANGLE BETWEEN TWO NON-ZERO VECTORS

If θ is the angle measure between two non-zero vectors $\vec{u}$ and $\vec{v}$ then $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|}$.

EXAMPLE

44

- a. Find the cosine of the angle between the vectors $\vec{u} = (2, 5)$ and $\vec{v} = (5, 2)$.
 b. Find the angle between the vectors $\vec{u} = (1, \sqrt{2})$ and $\vec{v} = (2, 2\sqrt{2})$.

Solution

- a. $\vec{u} \cdot \vec{v} = 2 \cdot 5 + 5 \cdot 2 = 20$
 $|\vec{u}| = \sqrt{2^2 + 5^2} = \sqrt{29}$
 $|\vec{v}| = \sqrt{5^2 + 2^2} = \sqrt{29}$
 $\cos \theta = \frac{20}{\sqrt{29} \cdot \sqrt{29}}$ by the formula, and so $\cos \theta = \frac{20}{29}$.
 b. $\vec{u} \cdot \vec{v} = 1 \cdot 2 + \sqrt{2} \cdot 2\sqrt{2} = 2 + 4 = 6$
 $|\vec{u}| = \sqrt{1^2 + 2} = \sqrt{3}$
 $|\vec{v}| = \sqrt{4 + 8} = \sqrt{12}$
 $\cos \theta = \frac{6}{\sqrt{3} \cdot \sqrt{12}} = \frac{6}{\sqrt{36}} = 1$

Therefore, $\theta = 0^\circ$.

Check Yourself 14

1. Find the dot product of $\vec{u} = (0, 3)$ and $\vec{v} = (-1, -2)$.
2. Find the length of $\vec{u} = (1, 3)$ using the dot product.
3. Find $(\vec{u} \cdot \vec{v}) \cdot \vec{w}$ given $\vec{u} = (0, 3)$, $\vec{v} = (1, 2)$, and $\vec{w} = (-2, -1)$.
4. Find the angle between $\vec{u} = (2, 1)$ and $\vec{v} = (1, 3)$.

Answers

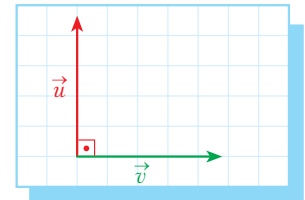
1. -6 2. $\sqrt{10}$ 3. $(-12, -6)$ 4. $\theta = 45^\circ$

2. Perpendicular and Parallel Vectors

Definition

perpendicular (orthogonal) vectors

Two non-zero vectors $\vec{u}$ and $\vec{v}$ are **perpendicular** if and only if the angle measure between them is 90° .



By the dot product theorem, $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta$. We can conclude that for two non-zero vectors $\vec{u}$ and $\vec{v}$, $\vec{u} \cdot \vec{v}$ is zero if and only if θ equals 90° .

Theorem

Two non-zero vectors $\vec{u}$ and $\vec{v}$ are perpendicular if and only if $\vec{u} \cdot \vec{v} = 0$.

EXAMPLE

45

Are the vectors $\vec{u} = (6, -2)$ and $\vec{v} = (1, 3)$ perpendicular?

Solution

$$\vec{u} \cdot \vec{v} = 6 \cdot 1 + (-2) \cdot 3 = 0$$

So the vectors are perpendicular.

EXAMPLE

46

Are the vectors $\vec{u} = (3, 5)$ and $\vec{v} = (2, -6)$ perpendicular?

Solution

$$\vec{u} \cdot \vec{v} = 3 \cdot 2 + 5 \cdot (-6) = 6 - 30 = -24. \text{ So } \vec{u} \text{ and } \vec{v} \text{ are not perpendicular.}$$

From the definition of parallel vectors we know that $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} = k \cdot \vec{v}$. We can write,

$$|\vec{u}| = |k| \cdot |\vec{v}| \quad (1)$$

$$\vec{u} \cdot \vec{v} = k \cdot \vec{v} \cdot \vec{v} = k \cdot |\vec{v}|^2 \quad (2)$$

$$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos \theta \quad (3).$$

From (1), (2) and (3),

$$|k| \cdot |\vec{v}| \cdot |\vec{v}| \cdot \cos \theta = k \cdot |\vec{v}|^2$$

$$\cos \theta = \frac{k}{|k|}$$

$$\cos \theta = 1 \text{ or } \cos \theta = -1$$

$$\theta = 0 \text{ or } \theta = 180.$$

From these results we can conclude that two vectors are parallel if and only if the angle measure between them is 0° or 180° .

Conclusion

Let θ be the angle measure between nonzero vectors $\vec{u}$ and $\vec{v}$.

Then $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}|$ or $\vec{u} \cdot \vec{v} = -|\vec{u}| \cdot |\vec{v}|$.

EXAMPLE

47

Determine two vectors in the plane which are perpendicular to $\vec{u} = (4, 2)$.

Solution

Let $\vec{v} = (v_1, v_2)$ be perpendicular to $\vec{u}$.

$$\vec{u} \cdot \vec{v} = 4 \cdot v_1 + 2 \cdot v_2 = 0$$

$$v_2 = -2 \cdot v_1$$

If $v_1 = t \in \mathbb{R}$, $v_2 = -2t$, then $\vec{v} = (t, -2t)$.

Let $t = 1$ or 3 :

$\vec{v} = (1, -2)$ and $\vec{v} = (3, -6)$ are both perpendicular to $\vec{u}$.

EXAMPLE

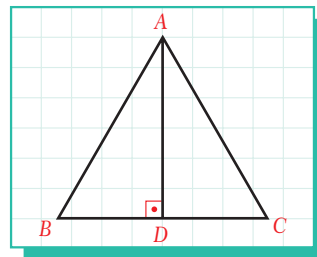
48

In an equilateral triangle ABC , D is the midpoint of BC . Find $\vec{AD} \cdot \vec{BC}$.

Solution

In an equilateral triangle, the median is also the altitude, as show in the diagram.

So $AD \perp BC$ and $\vec{AD} \cdot \vec{BC} = 0$.



EXAMPLE

49

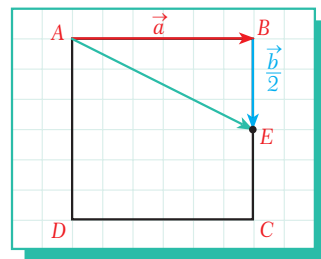
In a square $ABCD$, E is the midpoint of side BC and $|BC| = 4$ cm. Find $\vec{AE} \cdot \vec{AB}$.

Solution

$$\begin{aligned}\vec{AB} &= \vec{a} \\ \vec{AE} &= \vec{a} + \frac{\vec{b}}{2}\end{aligned}$$

$$\vec{AE} \cdot \vec{AB} = \vec{a} \cdot \left(\vec{a} + \frac{\vec{b}}{2} \right) = \vec{a}^2 + \frac{\vec{a} \cdot \vec{b}}{2} = \vec{a}^2$$

$$\text{Since } \vec{a} \perp \vec{b}, \vec{a}^2 = |\vec{a}|^2 = 4^2 = 16.$$



EXAMPLE

50

Find the area of the triangle with vertices $A(2, 3)$, $B(0, 1)$, $C(3, 2)$.

Solution

$$A(\widehat{ABC}) = \frac{|\vec{AH}| \cdot |\vec{BC}|}{2}$$

$$\vec{AH} = (x_0 - 2, y_0 - 3)$$

$$\vec{BC} = (3, 1)$$

$$\vec{AH} \cdot \vec{BC} = 3 \cdot (x_0 - 2) + (y_0 - 3) = 0$$

$$3x_0 + y_0 - 6 - 3 = 0$$

$$3x_0 + y_0 = 9$$

$$\vec{BH} = k \vec{HC}$$

$$\vec{BH} = (x_0, y_0 - 1)$$

$$\vec{HC} = (3 - x_0, 2 - y_0)$$

$$\frac{x_0}{3 - x_0} = \frac{y_0 - 1}{2 - y_0}$$

$$2x_0 - x_0y_0 = 3y_0 - 3 - x_0y_0 + x_0$$

$$x_0 = 3y_0 - 3$$

$$x_0 - 3y_0 = -3$$

$$3x_0 + y_0 = 9 \quad (1)$$

$$x_0 - 3y_0 = -3 \quad (2)$$

Let us multiply (1) by 3. Then,

$$9x_0 + 3y_0 = 27$$

$$+ \quad x_0 - 3y_0 = -3$$

$$\hline 10x_0 = 24$$

$$x_0 = 2.4 \text{ and } y_0 = 1.8.$$

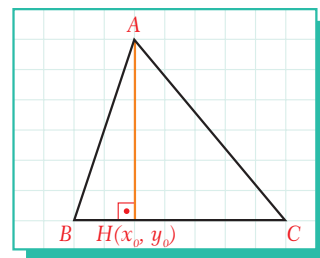
$$\vec{AH} = (0.4, -1.2)$$

$$|\vec{AH}| = \sqrt{0.16 + 1.44} = \sqrt{1.6} = 4 \cdot \frac{1}{\sqrt{10}}$$

$$\vec{BC} = (3, 1)$$

$$|\vec{BC}| = \sqrt{9 + 1} = \sqrt{10}$$

$$A(\widehat{ABC}) = \frac{1}{2} \cdot \frac{4}{\sqrt{10}} \cdot \sqrt{10} = 2$$



Theorem

triangle inequality

If $\vec{u}$ and $\vec{v}$ are vectors in the plane, then $|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|$. This property is called the **triangle inequality**.

Proof

By the property of the dot product, $|\vec{u} + \vec{v}|^2 = (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = \vec{u} \cdot \vec{u} + 2\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{v}$.

$$2\vec{u} \cdot \vec{v} \leq 2 \cdot |\vec{u}| \cdot |\vec{v}|$$

$$\{|\vec{u}|^2 + |\vec{v}|^2 + 2\vec{u} \cdot \vec{v}\} \leq \{|\vec{u}|^2 + |\vec{v}|^2 + 2|\vec{u}| \cdot |\vec{v}|\}$$

$|\vec{u} + \vec{v}|^2 \leq (|\vec{u}| + |\vec{v}|)^2$, since both $|\vec{u}| + |\vec{v}|$ and $|\vec{u} + \vec{v}|$ are non-negative.

Therefore, $|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|$.

Theorem

$\vec{u}$ and $\vec{v}$ are perpendicular in the plane if and only if $|\vec{u} + \vec{v}|^2 = |\vec{u}|^2 + |\vec{v}|^2$.

Proof

$$\begin{aligned} |\vec{u} + \vec{v}|^2 &= (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = \vec{u} \cdot \vec{u} + 2\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{v} = |\vec{u}|^2 + 2\vec{u} \cdot \vec{v} + |\vec{v}|^2 \\ &= |\vec{u}|^2 + |\vec{v}|^2, \text{ since } \vec{u} \cdot \vec{v} = 0. \end{aligned}$$

EXAMPLE

51

$|\vec{u}| = 3$, $|\vec{v}| = 4$, and the angle between $\vec{u}$ and $\vec{v}$ is 60° . Find $|2\vec{u} - 3\vec{v}|$.

Solution

$$\begin{aligned} |2\vec{u} - 3\vec{v}|^2 &= (2\vec{u} - 3\vec{v}) \cdot (2\vec{u} - 3\vec{v}) = 4\vec{u} \cdot \vec{u} + 9\vec{v} \cdot \vec{v} - 12\vec{u} \cdot \vec{v} \\ &= 4|\vec{u}|^2 + 9|\vec{v}|^2 - 12|\vec{u}||\vec{v}|\cos\theta \\ &= 4 \cdot 3^2 + 9 \cdot 4^2 - 12 \cdot 3 \cdot 4 \cdot \frac{1}{2} \\ &= 36 + 144 - 72 \\ &= 108. \end{aligned}$$

Now $|2\vec{u} - 3\vec{v}|^2 = 108$, and so $|2\vec{u} - 3\vec{v}| = \sqrt{108} = 6\sqrt{3}$.

EXAMPLE

52

Find an equation for the line passing through $A(-1, 3)$ which is perpendicular to $\vec{n} = (3, 5)$.

Solution

Let us choose a variable point $B(x, y)$:

$$\vec{AB} = (x + 1, y - 3).$$

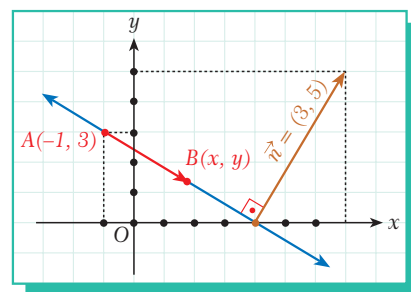
$\vec{AB}$ and $\vec{n}$ are perpendicular, so

$$\vec{AB} \cdot \vec{n} = 0.$$

$$3 \cdot (x + 1) + 5 \cdot (y - 3) = 0$$

$$3x + 3 + 5y - 15 = 0$$

$3x + 5y - 12 = 0$ is the required equation.



Check Yourself 15

- Find the value of a if $\vec{u} = (a, 1)$ is perpendicular to $\vec{v} = (3, 4)$.
- In the right triangle ABC , AC is the hypotenuse, $BH \perp AC$, and H lies on AC .
Find $(\vec{HC} + \vec{CB}) \cdot (\vec{AB} + \vec{BH})$.
- Find an equation for the line passing through $P(3, -1)$ which is perpendicular to $\vec{n} = (3, 1)$.

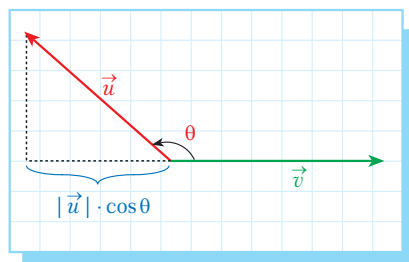
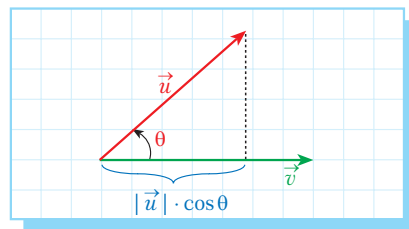
Answers

- $a = -\frac{4}{3}$
- 0
- $3x + y - 8 = 0$

C. COMPONENT OF $\vec{u}$ ALONG $\vec{v}$ (OPTIONAL)

The **scalar component of $\vec{u}$ along $\vec{v}$** (or the scalar component of $\vec{u}$ in the direction of $\vec{v}$) is given by $|\vec{u}| \cdot \cos \theta$, where θ is the angle measure between $\vec{u}$ and $\vec{v}$. We can interpret this concept geometrically as shown in the figures opposite. The scalar component of $\vec{u}$ along $\vec{v}$ is the magnitude of the portion of $\vec{u}$ that points in the direction of $\vec{v}$. We can see in the figures that when θ is acute, the scalar component of $\vec{u}$ along $\vec{v}$ is $|\vec{u}| \cdot \cos \theta$. When θ is obtuse, the scalar component of $\vec{u}$ along $\vec{v}$ is negative, and the length is $-|\vec{u}| \cdot \cos \theta$. As we know from the previous section,

$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|}$, and so we can write a formula to express the component of $\vec{u}$ along $\vec{v}$:



COMPONENT OF $\vec{u}$ ALONG $\vec{v}$

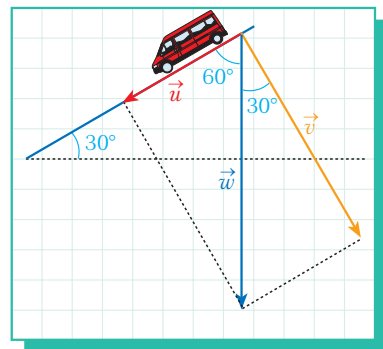
$$|\vec{u}| \cdot \cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|}$$

EXAMPLE

53

A bus weighing 6000 N is parked on a driveway that is inclined 30° to the horizontal, as shown in the figure.

- Find the magnitude of the force required to prevent the bus from rolling down the driveway.
- Find the magnitude of the force experienced by the driveway due to the weight of the bus.



Solution The bus exerts a force $\vec{w}$ of 6000 N directly downward.

We need to determine how much of this force causes the bus to roll down the driveway, and how much of it is exerted on the surface of the driveway. This means that we must resolve $\vec{w}$ into its components.

- a. The magnitude of the part of the force $\vec{w}$ that causes the bus to roll down the driveway is $|\vec{u}| = \text{scalar component of } \vec{w} \text{ along } \vec{u} = 6000 \cdot \cos 60^\circ = 3000$.

Thus, the force needed to prevent the bus from rolling down the driveway is 3000 N.

- b. The magnitude of the force exerted by the bus on the driveway is $|\vec{v}| = \text{scalar component of } \vec{w} \text{ along } \vec{v} = 6000 \cdot \cos 30^\circ = 3000\sqrt{3}$ N.

Therefore, the force exerted by the bus on the driveway is $3000\sqrt{3}$ N.

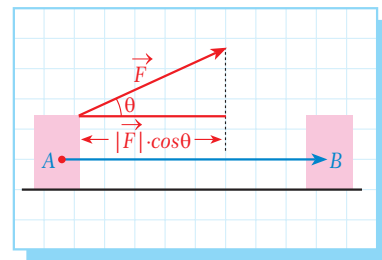
EXAMPLE 54 Find the scalar component of $\vec{u}$ along $\vec{v}$ if $\vec{u} = (1, 4)$ and $\vec{v} = (-2, 1)$.

Solution By the formula, the scalar component of $\vec{u}$ along $\vec{v}$ is $\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} = \frac{1 \cdot (-2) + 4 \cdot 1}{\sqrt{4 + 1}} = \frac{2}{\sqrt{5}}$.

Units for work and force	
force	work
newton	newton-meter
N	N·m
$1 \text{ N} = 1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$	$1 \text{ J} = 1 \text{ N} \cdot \text{m}$

In physics, the work W done by a constant force $\vec{F}$ when the application of $\vec{F}$ points in the direction of displacement $\vec{AB}$ is defined by the formula

$$\begin{aligned} W &= (|\vec{F}| \cdot \cos \theta) \cdot |\vec{AB}| = |\vec{F}| \cdot |\vec{AB}| \cdot \cos \theta \\ &= \vec{F} \cdot \vec{AB} \quad (\vec{AB} = \vec{D}) \\ &= \vec{F} \cdot \vec{D}. \end{aligned}$$



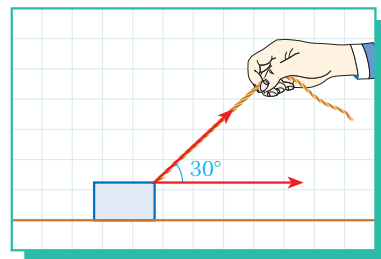
EXAMPLE 55 A force is given by $\vec{F} = (2, 3)$ N and an object moves from the point $A(1, 3)$ to $C(4, 7)$, where each unit is 1 m. Find the work done in newton-meters.

Solution The displacement vector is $\vec{D} = (4 - 1, 7 - 3) = (3, 4)$.

Therefore, the work done is $W = \vec{F} \cdot \vec{D} = (2, 3) \cdot (3, 4) = 2 \cdot 3 + 3 \cdot 4 = 18 \text{ N} \cdot \text{m}$.

EXAMPLE**56**

A student pulls a box horizontally across a table by exerting a force of 20 N on the handle. If the handle makes an angle of 30° with the horizontal, find the work done in moving the box 100 m.



Solution Let us choose a coordinate system with the origin at the initial position of the box. That is, the initial point of the displacement is $P(0, 0)$. The terminal point is $Q(100, 0)$. So the displacement vector is $\vec{D} = (100, 0)$.

The force on the handle is $\vec{F} = (20 \cdot \cos 30^\circ) \vec{i} + (20 \cdot \sin 30^\circ) \vec{j} = 10\sqrt{3} \cdot \vec{i} + 10\vec{j}$.

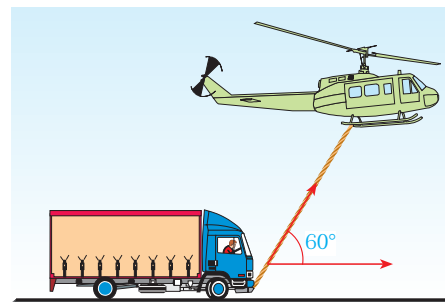
So, the work done is

$$W = \vec{F} \cdot \vec{D} = (10\sqrt{3} \vec{i} + 10\vec{j}) \cdot (100\vec{i}) = 100 \cdot 10\sqrt{3} = 1000\sqrt{3} \text{ N}\cdot\text{m}.$$

Check Yourself 16

1. Find the scalar component of $\vec{u}$ along $\vec{v}$ if $\vec{u} = (4, 1)$ and $\vec{v} = (-2, 2)$.

2. A helicopter is pulling a truck horizontally due east. The tension exerted by the rope connecting the truck and the helicopter is 10^3 newtons. If the angle between the connecting rope and the horizontal path is 60° , how many newton-meters will it take to pull the truck 50 kilometers?

**Answers**

1. $\frac{-3}{\sqrt{2}}$ 2. 25 000 N·m

Projectile Motion in Sport

Look at the problem:

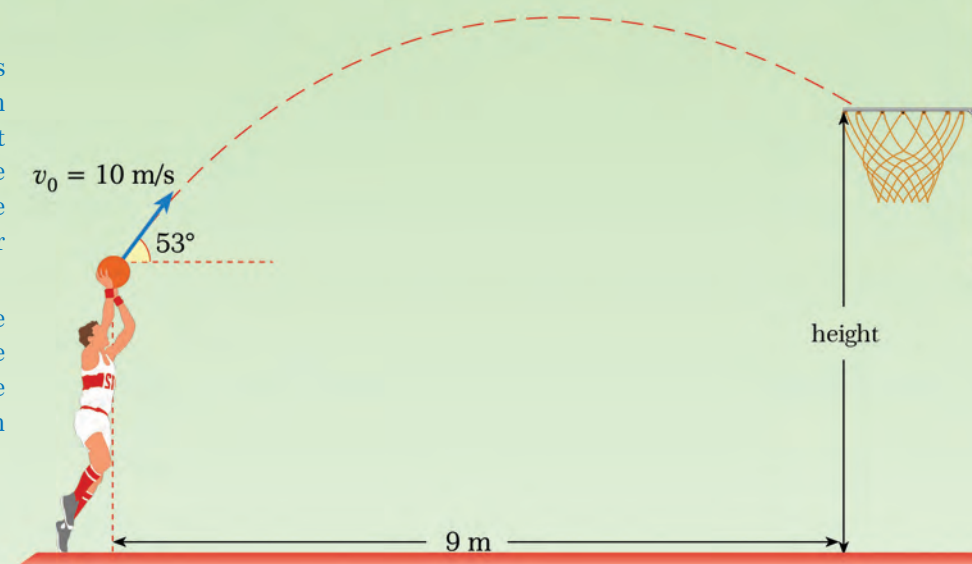
A basketball player is 1.85 m tall. He throws a basketball into a basket from 0.45 m above his head, with a velocity of 10 m/s at an angle of 53° to the horizontal. If the player is 9 m away from the basket when he throws the ball, how many seconds later does the ball pass through the basket?

We can solve this problem by using the horizontal component of the velocity of the ball and the distance it travels in the horizontal direction. Let t be the time in seconds. Then

$$x = v_0 \cos 53^\circ \cdot t \quad (\text{resolving a vector})$$

$$9 = 10 \cdot 0.6t, \quad (\cos 53^\circ \cong 0.6)$$

$$\text{thus } t = 1.5 \text{ s.}$$

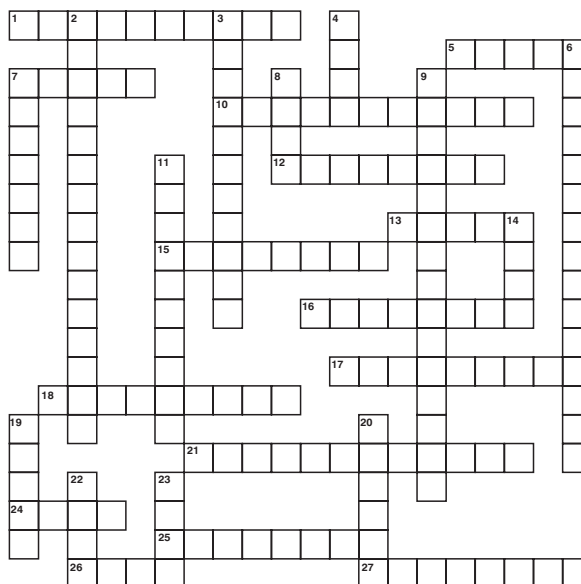


In the picture, a soccer player has kicked the football. The maximum height of the ball and the maximum height of the goalkeeper's hand are the same, but the ball passes over the goalkeeper. Why can't the goalkeeper catch the ball?

The reason is timing. While the goalkeeper is jumping up, the ball is already at its maximum height.



Project: Investigate other projectile motion problems.



ACROSS

1. A vector with length zero.
5. An undefined concept in geometry which has no shape or size.
7. Being the main or most important part.
10. Being parallel.
12. The coordinate representing the position of a point along a line perpendicular to the y -axis in a plane.
13. A thin straight stick with a point, sometimes used to shoot things.
15. A mathematical statement that establishes the equality of two expressions.
16. Any of the four areas into which a plane is divided by the reference axes in a rectangular coordinate system, designated first, second, third, and fourth, counting counterclockwise from the area in which both coordinates are positive.
17. One of the two horizontal or vertical vectors whose sum is equal to a given vector.
18. In a triangle, the formula $\cos A = (b^2 + c^2 - a^2)/2bc$.
21. Vectors which have the same direction and length.
24. To arrange in sets of two.
25. The act or process of adding.
26. The direction 90° clockwise from north.
27. Either of two points marking the end of a line segment.

DOWN

2. A single vector that is the equivalent of a set of vectors.
3. The measure of heat of a body or environment.
4. The principal structural member of a ship.
6. The solution of an equation in which every variable is equal to zero.
7. An acute angle measured from due north or due south.
8. The size of a flat surface, calculated mathematically.
9. The property that states: if $a, b \in A$, then $a \cdot b \in A$ for an operation.
11. A point whose position is constant.
14. The direction 270° clockwise from due north and directly opposite east.
19. A picture such as a pie chart or bar graph, used to illustrate quantitative relationships.
20. To give the meaning of (a word or idea).
22. An undefined concept in geometry that describes a set of points along a path.
23. A relatively small, usually open craft used by fishermen.

EXERCISES 3

A. Dot Product

1. Find the dot product of each pair of vectors.

$$\begin{array}{ll} \text{a. } \vec{u} = (-2, 1) & \vec{v} = (3, 6) \\ \text{b. } \vec{u} = -7\vec{i} + 24\vec{j}, & \vec{v} = 3\vec{i} - \vec{j} \\ \text{c. } \vec{u} = \sqrt{3}\vec{i} + \vec{j}, & \vec{v} = 3\vec{i} - \vec{j} \\ \text{d. } \vec{u} = \vec{i} - \vec{j}, & \vec{v} = 2\vec{i} \\ \text{e. } \vec{u} = -19\vec{j}, & \vec{v} = 16\vec{j} \end{array}$$

2. Find the following quantities given $\vec{u} = \vec{i} + 2\vec{j}$, $\vec{v} = -3\vec{i} + \vec{j}$, and $\vec{w} = 4\vec{i} + 3\vec{j}$.

$$\begin{array}{l} \text{a. } \vec{u} \cdot \vec{w} + \vec{u} \cdot \vec{w} \\ \text{b. } \vec{w} \cdot (\vec{u} + \vec{v}) \\ \text{c. } (\vec{u} + \vec{w}) \cdot (\vec{u} - \vec{w}) \\ \text{d. } 2 \cdot (\vec{u} \cdot \vec{v}) \\ \text{e. } (\vec{w} - \vec{v}) \cdot (\vec{w} + \vec{u}) \\ \text{f. } (3\vec{u} - 2\vec{v}) \cdot (\vec{u} + 2\vec{v}) \end{array}$$

3. $\vec{u}$, $\vec{v}$ and $\vec{w}$ are unit vectors such that $\vec{u} + \vec{v} + \vec{w} = 0$. Find $\vec{u} \cdot \vec{v} + \vec{v} \cdot \vec{w} + \vec{w} \cdot \vec{u} = 0$.

4. Evaluate the following quantities if the angle measure between $\vec{u}$ and $\vec{v}$ is 60° and $|\vec{u}| = 4$, $|\vec{v}| = 3$.

$$\begin{array}{ll} \text{a. } \vec{u} \cdot \vec{v} & \text{b. } |\vec{u} + \vec{v}|^2 \\ \text{c. } |\vec{u} - \vec{v}|^2 & \text{d. } (3\vec{u} + 2\vec{v}) \cdot (\vec{u} + 2\vec{v}) \end{array}$$

5. Evaluate $|\vec{u} + \vec{v}|$ given $|\vec{u}| = 13$, $|\vec{v}| = 5$, and $|\vec{u} - \vec{v}| = 12$.

B. Angle Between Two Vectors

6. Determine whether the given vectors are perpendicular or not.

$$\begin{array}{ll} \text{a. } \vec{u} = (4, 6), & \vec{v} = (3, -2) \\ \text{b. } \vec{u} = (-5, 0), & \vec{v} = (0, 4) \\ \text{c. } \vec{u} = 3\vec{i}, & \vec{v} = -\vec{j} \end{array}$$

$$\text{d. } \vec{u} = 2\vec{i} - 8\vec{j}, \quad \vec{v} = 12\vec{i} + 3\vec{j}$$

$$\text{e. } \vec{u} = 3\vec{i} - 4\vec{j}, \quad \vec{v} = 8\vec{i} - 6\vec{j}$$

$$\text{f. } \vec{u} = 4\vec{i}, \quad \vec{v} = 3\vec{i}$$

7. Find the measure of the interior angles of the triangle ABC with vertices $A(1, 3)$, $B(1, 2)$, and $C(2, 2)$.

8. Find two perpendicular vectors to each given vector.

$$\text{a. } \vec{a} = 3\vec{i} - \vec{j} \quad \text{b. } \vec{b} = -8\vec{i} - 6\vec{j} \quad \text{c. } \vec{c} = \vec{i} + 2\vec{j}$$

9. For what values of t are $\vec{u} = 4\vec{i} - 5\vec{j} \cdot t$ and $\vec{v} = 3\vec{i} - 2\vec{j}$ perpendicular?

10. $|\vec{u}| = 5$ and $|\vec{v}| = 3$ are given. For what values of t are $\vec{u} + t\vec{v}$ and $\vec{u} - t\vec{v}$ perpendicular?

11. For what values of t do $\vec{a} = t\vec{i} + \vec{j}$ and $\vec{b} = \vec{i} + t\vec{j}$ have angle measure 120° between them?

12. Show that the equality $|\vec{u} + \vec{v}| = |\vec{u} - \vec{v}|$ holds when $\vec{u}$ and $\vec{v}$ are orthogonal.

13. Verify the equality $|\vec{u} + \vec{v}|^2 + |\vec{u} - \vec{v}|^2 = 2(|\vec{u}|^2 + |\vec{v}|^2)$.

14. In a rhombus $ABCD$, one side measures 6 cm and E , F are the midpoints of sides AD and DC respectively. The angle measure between sides AD and DC is 120° . Find the quantity $\vec{BE} \cdot \vec{AF}$.

15. In a rectangle $ABCD$, $\vec{DC} = 3\vec{AD}$ and point E is on DC . Find the quantity $\vec{AE} \cdot \vec{BE}$ given $\vec{DE} = 2\vec{EC}$ and $|\vec{AD}| = 3$ cm.

16. Show that the inequality $|\vec{u} + \vec{v}| > |\vec{u} - \vec{v}|$ holds when the angle measure between $\vec{u}$ and $\vec{v}$ is less than 90° .

17. Find two unit vectors that make angles of 45° with $-\vec{i} + \vec{j}$.

18. Let $\vec{u}$ and $\vec{v}$ be vectors and let α be a scalar. Verify the given properties.

a. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$

b. $(\alpha \cdot \vec{u}) \cdot \vec{v} = \alpha \cdot (\vec{u} \cdot \vec{v}) = \vec{u} \cdot (\alpha \cdot \vec{v})$

19. Write the equation of the line passing through $A(-1, -1)$ which is perpendicular to $\vec{u} = (3, 4)$.

20. Show that the diagonals of a rhombus are perpendicular using vectors.

21. Find the area of a triangle with vertices $A(2, 2)$, $B(0, 2)$, and $C(1, 4)$.

22. Find the area of a rhombus with vertices $A(2, 0)$, $B(-3, 3)$, $C(-8, 0)$, and $D(-3, -3)$.

23. Find the area of a rectangle with vertices $A(3, 2)$, $B(9, 2)$, $C(9, 5)$, and $D(3, 5)$.

24. For any vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ prove that $(\vec{u} \cdot \vec{w}) \cdot \vec{v} - \vec{u} \cdot (\vec{v} \cdot \vec{w})$ is perpendicular to $\vec{w}$.

25. Show that the altitudes of an acute-angled triangle are concurrent using vectors.

26. Find the distance of $P(x_0, y_0)$ from the line $ax + by + c = 0$ using vectors.

27. Find the area of a parallelogram with vertices $A(2, -1)$, $B(6, 0)$, $C(8, 3)$, and $D(4, 2)$.

C. Component of $\vec{u}$ Along $\vec{v}$ (Optional)

28. Find the scalar component of $\vec{u}$ along $\vec{v}$ for each pair of vectors.

a. $\vec{u} = (4, 6)$, $\vec{v} = (3, -4)$

b. $\vec{u} = 7\vec{i} - 24\vec{j}$, $\vec{v} = \vec{j}$

29. Find the work done by the force F in moving an object from P to Q given each set of data.

a. $\vec{F} = 5\vec{i} - 4\vec{j}$, $P(3, 8)$, $Q(0, 0)$

b. $\vec{F} = 40\vec{i} - 5\vec{j}$, $P(1, -1)$, $Q(20, 1)$

30. A car is on a driveway inclined 30° to the horizontal. If the car weighs 28 N, find the force that prevents the car from rolling down the driveway.

31. A windsurfing board has its sail inclined in the direction $N 50^\circ W$ and its direction is due west. The wind is blowing into the sail in the direction $N 20^\circ W$ with a force of 20 N, as shown in the figure.

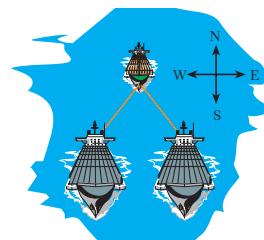


a. Find the effective force of the wind on the sail.

[Hint: Find the components of the wind parallel to the sail and perpendicular to the sail.]

b. If the front of the board is turned due north, find the effective force of the wind that drives the board forward.

32. Two boats are pulling a ship, as shown. One boat pulls with a force of $\sqrt{2} \cdot 10^4$ N in the direction $S 45^\circ W$, and the other boat pulls with a force of $3.4 \cdot 10^4$ N in the direction $S 75^\circ E$.



a. Find the resultant force on the ship as a vector.

b. Find the magnitude of the force ($\cos 75^\circ \approx 0.258$, $\sin 75^\circ \approx 0.965$).

CHAPTER SUMMARY

- A line segment with direction is called a directed line segment.
- A vector in the plane is a directed line segment.

• Addition of Two Vectors Geometrically

For two vectors $\vec{PQ}$ and $\vec{QR}$ in a plane, $\vec{PQ} + \vec{QR}$ is the sum of $\vec{PQ}$ and $\vec{QR}$. There are two ways to add vectors geometrically: the polygon method and the parallelogram method.

• The Polygon Method

In this method we draw the first vector. Then we place the initial point of the second vector at the terminal point of the first, the initial point of the third vector at the terminal point of the second, and so on until we place the initial point of the n^{th} vector at the terminal point of the $(n - 1)^{\text{th}}$ vector. The vector whose initial point is the initial point of the first vector and whose terminal point is the terminal point of the last vector is the resultant (sum) vector.

• The Parallelogram Method

In this method we draw the first vector, and then draw the second vector with its initial point at the initial point of the first vector. We make a parallelogram by drawing two additional sides, each passing through the terminal point of one of the vectors and parallel to the other vector. The sum is drawn along the diagonal from the common initial point to the intersection of the two lines.

• Multiplication of a Vector by a Scalar

For a real number a and a vector $\vec{u}$:

1. If $a > 0$, then the vector $a \cdot \vec{u}$ has the same direction to $\vec{u}$ and the length $|a \cdot \vec{u}| = a \cdot |\vec{u}|$.
2. If $a < 0$, then the vector $a \cdot \vec{u}$ has the opposite direction to $\vec{u}$ and the length $|a \cdot \vec{u}| = |a| \cdot |\vec{u}|$.
3. If $a = 0$, then $a \cdot \vec{u} = \vec{0}$.

• Parallel Vectors

Let $\vec{a}$ and $\vec{b}$ be two vectors. $\vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{a} = k \cdot \vec{b}$ where $k \neq 0$.

• Non-Parallel Vector

If $\vec{a}$ is not parallel to $\vec{b}$, then $h \cdot \vec{a} = k \cdot \vec{b}$ when $h = k = 0$.

• Components of a Vectors

A vector in a plane has two components, called the horizontal and vertical components of the vector.

Let $|\vec{a}| = (a_1, a_2)$ be a vector, then $|\vec{a}| = \sqrt{a_1^2 + a_2^2}$.

• Equal Vectors

Two vectors are equal if and only if their corresponding component vectors are equal.

In other words, for $\vec{a} = (a_1, a_2)$ and $\vec{b} = (b_1, b_2)$, then $\vec{a} = \vec{b}$ if and only if $a_1 = b_1$ and $a_2 = b_2$.

• Standard Base Vectors

$\vec{i} = (1, 0)$ and $\vec{j} = (0, 1)$ are called the standard base vectors.

• Linear Combination of Vectors

Let $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_k$ be vectors in a plane and let $c_1, c_2, \dots, c_k$ be scalars. $\vec{v} = c_1 \cdot \vec{u}_1 + c_2 \cdot \vec{u}_2 + \dots + c_k \cdot \vec{u}_k$ is called a linear combination of vectors.

• Dot Product

The dot product of $\vec{u} = (u_1, u_2)$ and let $\vec{v} = (v_1, v_2)$ is the scalar quantity $\vec{u} \cdot \vec{v} = u_1 \cdot v_1 + u_2 \cdot v_2$.

• Angle Between Two Vectors

Let θ be the angle between two non-zero vectors $\vec{u}$ and $\vec{v}$.

1. $\vec{u} \parallel \vec{v}$ if and only if $\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}|$ or $\vec{u} \cdot \vec{v} = -|\vec{u}| \cdot |\vec{v}|$ because $\theta = 0^\circ$ or $\theta = 180^\circ$.
2. $\vec{u}$ and $\vec{v}$ are perpendicular if and only if $\vec{u} \cdot \vec{v} = 0$.

Concept Check

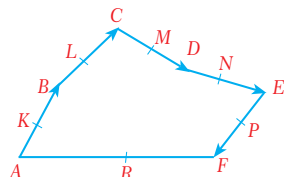
1. Can we use directed line segments in traffic? Give an example.
2. What is the difference between a vector and a scalar?
3. Is it possible to add a hundred vectors in a plane using the polygon method or the parallelogram method?
4. How do you change the direction of a vector using a real number?
5. a. Draw a diagram to show how to add two vectors.
b. Draw a diagram to show how to subtract two vectors.
6. Can you equalize two non-parallel vectors using multiplication by two real numbers?
7. In a plane, how many components do you need to resolve a vector?
8. How many standard base vectors are there in the plane?
9. What is the difference between the inner product and dot product of two vectors?
10. How do you use the dot product to find the angle measure between two vectors?
11. How do you use the dot product to determine whether two vectors are parallel or not?
12. How do you use the dot product to determine whether two vectors are perpendicular or not?

CHAPTER REVIEW TEST 1

1. In triangle ABC , G is the centroid. Describe $\vec{AG} + \vec{GC} + \vec{CA}$.

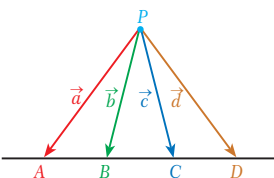
A) $\vec{0}$ B) $\vec{AG}$ C) $\vec{BG}$ D) $2\vec{CA}$ E) $\vec{AC}$

2. In the figure, K, L, M, N, P, R are the midpoints of the sides $ABCDEF$ respectively. Describe $\vec{KB} + \vec{LC} + \vec{MD} + \vec{NE} + \vec{PF}$.



A) $\vec{AR}$ B) $\vec{RA}$ C) $\vec{FK}$ D) $\vec{FA}$ E) $\vec{PA}$

3. In the figure, $|AB| = |BC| = |CD|$ and the vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are given. Which one of the following is the linear combination of $\vec{c}$ in terms of $\vec{a}$ and $\vec{d}$?



A) $\frac{\vec{a} + \vec{d}}{2}$ B) $\frac{2\vec{a} + \vec{d}}{2}$ C) $\frac{2\vec{a} + \vec{d}}{3}$
D) $\frac{\vec{a} + 2\vec{d}}{3}$ E) $\frac{\vec{a} + 2\vec{d}}{2}$

4. $\vec{AD} = \frac{1}{4}\vec{AB}$ and $\vec{BE} = \frac{1}{2}\vec{BC}$ are given. Express $\vec{DE}$ in terms of $\vec{AB}$ and $\vec{BC}$.

A) $\frac{3}{4}\vec{AB} + \frac{1}{2}\vec{BC}$ B) $\frac{1}{4}\vec{AB} + \frac{1}{2}\vec{BC}$
C) $\frac{3}{4}\vec{AB} - \frac{1}{2}\vec{BC}$ D) $\frac{1}{4}\vec{AB} - \frac{1}{2}\vec{BC}$
E) $\frac{1}{2}\vec{BC} - \frac{3}{4}\vec{AB}$

5. $(-1, 4) + \vec{v} = (4, 5)$ is given. Find $\vec{v}$.

A) (3, 9) B) (-5, -1) C) (5, 1)
D) (1, 5) E) (9, 3)

6. Find the coordinates of B if $\vec{AB} = (7, 3)$ and $A(2, -1)$.

A) (9, -4) B) (9, -2) C) (9, 4)
D) (-9, -4) E) (9, 2)

7. $\vec{u} = -2\vec{i} + 3\vec{j}$ and $\vec{v} = \vec{i} - 2\vec{j}$ are given. Find $3\vec{u} - 2\vec{v}$.

A) $4\vec{i} + 5\vec{j}$ B) $8\vec{i} + 5\vec{j}$
C) $-8\vec{i} + 13\vec{j}$ D) $-4\vec{i} + 13\vec{j}$
E) $4\vec{i} + 13\vec{j}$

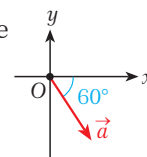
8. Find the unit vector with opposite direction to $\vec{AB}$ given $A(6, 3)$, and $B(2, 6)$.

A) $(\frac{4}{5}, \frac{3}{5})$ B) $(\frac{4}{5}, -\frac{3}{5})$ C) $(-\frac{4}{5}, \frac{3}{5})$
D) $(\frac{4}{7}, -\frac{3}{7})$ E) $(-\frac{4}{5}, -\frac{3}{5})$

9. $\vec{u} = \vec{i} - 5\vec{j}$ and $\vec{v} = 2\vec{i} + 3\vec{j}$ are given. Find $3\vec{u} + 2\vec{v}$.

A) $3\vec{i} - 2\vec{j}$ B) $5\vec{i} - \vec{j}$ C) $7\vec{i} - 9\vec{j}$
D) $2\vec{i} + 3\vec{j}$ E) $8\vec{i} - \vec{j}$

10. Describe the unit vector $\vec{a}$ in the figure.



A) $\frac{\sqrt{3}}{2}\vec{i} + \frac{1}{2}\vec{j}$ B) $-\frac{\sqrt{3}}{2}\vec{i} + \frac{1}{2}\vec{j}$
C) $\frac{1}{2}\vec{i} - \frac{\sqrt{3}}{2}\vec{j}$ D) $\frac{1}{2}\vec{i} + \frac{\sqrt{3}}{2}\vec{j}$
E) $\frac{\sqrt{3}}{2}\vec{i}$

11. For how many values of $m > 0$ are $\vec{a} = (2, 1 - m)$ and $\vec{b} = (m + 1, -4)$ parallel?

A) 1 B) 2 C) 3 D) 4 E) 5

12. Which one of the following is true for vectors $\vec{a} = (3, -6)$, $\vec{b} = (-6, 7)$ and $\vec{c} = (9, -13)$?

A) $\vec{a} = 4\vec{b} + 3\vec{c}$ B) $\vec{a} = 3\vec{b} + 4\vec{c}$

C) $\vec{a} = 4\vec{b} - 3\vec{c}$ D) $\vec{a} = 3\vec{b} - 4\vec{c}$

E) $\vec{a} = \vec{b} + \vec{c}$

13. Find the measure of the angle between the unit vectors $\vec{a}$ and $\vec{b}$ in degrees if their dot product is $\frac{1}{2}$.

A) 45° B) 60° C) 120° D) 135° E) 150°

14. $A(2, 5)$, $B(-1, 3)$, $C(m, 6)$, and $\vec{AB} \perp \vec{BC}$ are given. Find m .

A) 3 B) 2 C) 1 D) -2 E) -3

15. Which one of the following is perpendicular to $\vec{v} = (3, -4)$?

A) $(-\frac{4}{5}, -\frac{3}{5})$ B) $(4, -3)$ C) $(\frac{3}{5}, \frac{4}{5})$

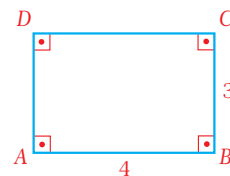
D) $(-\frac{3}{5}, -\frac{4}{5})$ E) $(-\frac{4}{5}, \frac{3}{5})$

16. $ABCD$ is the rectangle in the figure.

Find $\vec{AB} \cdot \vec{AC}$ if

$|AB| = 4$ and

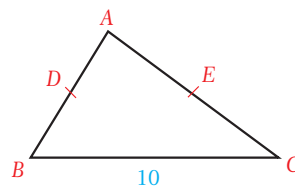
$|BC| = 3$.



A) 9 B) 12 C) 15 D) 16 E) 20

17. In the figure, D and E are the midpoints of the sides of triangle ABC .

Find $(\vec{AD} - \vec{AE}) \cdot \vec{BC}$.



A) 100 B) 150 C) 0 D) -50 E) -100

18. G is the centroid of a triangle with vertices $A(1, 1)$, $B(-4, 2)$, and $C(0, 6)$. Find the scalar product of $\vec{AB} \cdot \vec{BG}$.

A) -8 B) -14 C) 0 D) 1 E) 12

19. A triangle ABC has vertices $A(1, 1)$, $B(-4, 2)$, and $C(0, 6)$. Find the scalar component of $\vec{AB}$ along $\vec{BC}$.

A) $\frac{4}{\sqrt{2}}$ B) $-\frac{4}{\sqrt{2}}$ C) $\frac{3}{4}$ D) $-\frac{3}{4}$ E) $\frac{3}{2}$

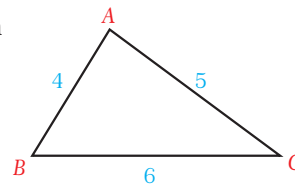
20. ABC is a triangle with

$|AB| = 4$,

$|AC| = 5$,

$|BC| = 6$. Find

$\vec{AC} \cdot (\vec{AB} + \vec{BC})$.



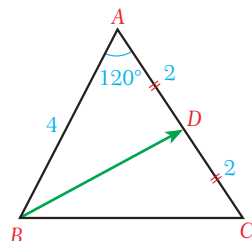
A) 20 B) 24 C) 25 D) 30 E) 50

CHAPTER REVIEW TEST 2

1. $|\vec{a}| = 3$, $|\vec{b}| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is 60° . Find $|\vec{a} + \vec{b}|$.

A) 7 B) 8 C) 9 D) 10 E) 11

2. In the figure,
 $|AB| = |AC| = 4$,
 $m\angle A = 120^\circ$, and
 $|AD| = |DC|$.
 Find $\vec{BC} \cdot \vec{BD}$.



A) 48 B) 36 C) 30 D) 24 E) 18

3. Given $|\vec{a}| = 12$, $|\vec{b}| = 5$, and $|\vec{a} - \vec{b}| = 8$, find $\cos(\vec{a}, \vec{b})$.

A) $\frac{3}{8}$ B) $\frac{7}{8}$ C) $\frac{3}{4}$ D) $\frac{5}{13}$ E) $\frac{12}{13}$

4. In an equilateral triangle ABC , $D \in [AC]$, $|BC| = 6$, and $|CD| = 2$. Find $\vec{BC} \cdot \vec{BD}$.

A) 30 B) 24 C) 20 D) 18 E) 12

5. $|\vec{a}| = 7$, $|\vec{b}| = 10$, and $|\vec{a} + \vec{b}| = \sqrt{73}$ are given. Find $|\vec{a} - \vec{b}|$.

A) 3 B) 8 C) 13 D) 15 E) 17

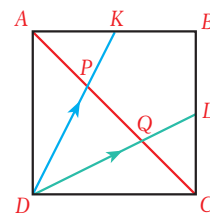
6. In the right triangle ABC , $m\angle A = 90^\circ$, $D \in [AC]$, and $|AB| = k$. Find $\vec{BA} \cdot \vec{BD}$.

A) k B) k^2 C) $\frac{k^2}{2}$ D) $\frac{k}{2}$ E) $2k^2$

7. What is the angle between $\vec{a} = (1, \sqrt{3})$ and $\vec{b} = (-4, -4\sqrt{3})$ in degrees?

A) 60° B) 120° C) 135° D) 150° E) 180°

8. A square $ABCD$ has side 3 cm, and K, L are the midpoints of sides AB and BC respectively. P and Q trisect AC as shown in the figure. Find $\vec{DP} \cdot \vec{PQ}$.



A) 0 B) 1 C) -1 D) -2 E) 3

9. $\vec{a}$, $\vec{b}$, and $\vec{c}$ are three vectors such that $\vec{b} = \vec{a} - 2\vec{c}$, $\vec{b} \perp \vec{c}$, and $|\vec{a}| = 4|\vec{c}|$. Find the angle between $\vec{a}$ and $\vec{c}$ in degrees.

A) 30° B) 45° C) 60° D) 75° E) 90°

10. Which one of the following is false for two vectors $\vec{a}$, $\vec{b}$ and a scalar $k \in \mathbb{R}$?

A) If $\vec{a} \perp \vec{b}$, $\vec{a} \cdot \vec{b} = 0$.
 B) If $\vec{a} \parallel \vec{b}$, $\vec{b} \cdot \vec{a} = 0$.
 C) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
 D) $(k \cdot \vec{a} + k \cdot \vec{b}) = k \cdot (\vec{a} + \vec{b})$
 E) $(k \cdot \vec{a} \cdot k \cdot \vec{b}) = k^2 \cdot \vec{a} \cdot \vec{b}$

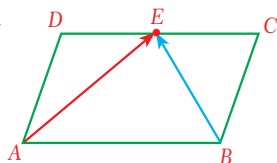
11. In the figure, $ABCD$ is a

$$|BC| = 1,$$

$$|DC| = 2,$$

$$|DE| = |EC|. \text{ Find}$$

$$(\vec{BC} + \vec{CE}) \cdot (\vec{AD} + \vec{DE}).$$



- A) 0 B) 1 C) 2 D) 3 E) 4

12. In an equilateral triangle ABC , D and E are the midpoints of $[AC]$ and $[BC]$ respectively.

$$\text{Find } \vec{DC} \cdot (\vec{BA} + \vec{AE}).$$

- A) $|\vec{AB}|^2$ B) $|\vec{EB}|^2$ C) $-\frac{1}{2} \cdot |\vec{DC}|^2$
D) $\frac{1}{2} \cdot |\vec{DC}|^2$ E) $\frac{1}{2} \cdot |\vec{EB}|^2$

13. In an equilateral triangle ABC with side 1 cm, what is $\vec{BC} \cdot \vec{CA}$?

- A) $\frac{1}{2}$ B) $-\frac{1}{2}$ C) $-\frac{\sqrt{3}}{2}$ D) -2 E) $\frac{\sqrt{3}}{2}$

14. For what values of k are the vectors $\vec{a} = (12k, 9)$ and $\vec{b} = (-4, 3)$ linearly dependent?

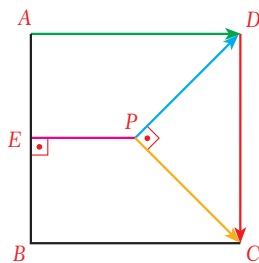
- A) -3 B) -2 C) -1 D) -0 E) 1

15. In a square $ABCD$, the side is 4 cm,

$$[AB] \perp [EP], [PD] \perp [PC],$$

$$\text{and } |AE| = |EB|.$$

$$\text{Find } \vec{PE} \cdot (\vec{PC} + \vec{PD}).$$



- A) -16 B) -12 C) -10 D) -8 E) -6

16. In a triangle ABC , $\vec{AB} = (-4, 2a)$, $\vec{AC} = (a, -4)$ and the length of $\vec{BC}$ is 10 cm. Find a possible value of a .

- A) 1 B) 2 C) 3 D) 4 E) 5

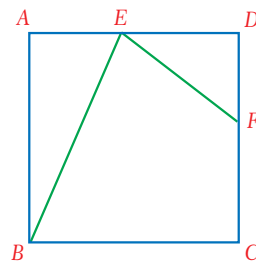
17. In a square $ABCD$

with side 2 cm,

$$|AE| = |ED| \text{ and}$$

$$|DF| = |FC|.$$

$$\text{Find } \vec{EF} \cdot (\vec{EA} + \vec{AB}).$$



- A) 1 B) 2 C) -2 D) -1 E) 0

18. $\vec{a} + \vec{b} = (1, 3)$ and $2\vec{a} + \vec{b} = (4, 6)$ are given. Find the measure of the angle between $\vec{a}$ and $\vec{b}$.

- A) 30° B) 45° C) 90° D) 135° E) 150°

19. In a triangle ABC , $\vec{AB} = (-2, -5)$ and $\vec{AC} = (2, -2)$. Find the length of $\vec{BC}$.

- A) 1 B) 2 C) 3 D) 4 E) 5

20. $\vec{a} = (12, -5)$, and $\vec{b} = (3, 4)$ are given. Find $\sin(\angle \vec{a}, \vec{b})$.

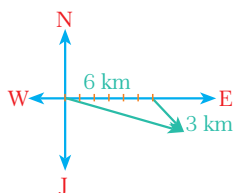
- A) $\frac{5}{13}$ B) $\frac{12}{13}$ C) $\frac{63}{65}$ D) $\frac{10}{13}$ E) $\frac{4}{13}$

ANSWERS TO EXERCISES

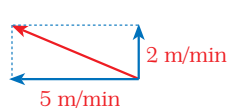
EXERCISES 1

2. an infinite number

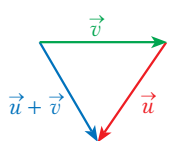
3. a.



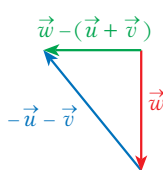
b.



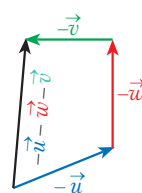
4. a.



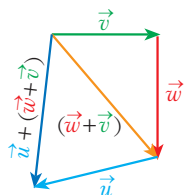
b.



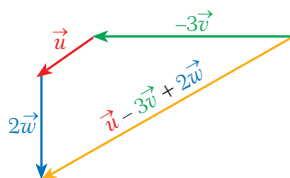
c.



d.



e.



$$5. \vec{OK} = \frac{1}{2} (\vec{OA} + \vec{OB}) \quad 6. \vec{AD} + \vec{AE} = \vec{AB} + \vec{AC}$$

7. look at the fifth property of vector addition

$$8. \text{ use } \vec{AG} + \vec{BG} + \vec{CG} = \vec{0}$$

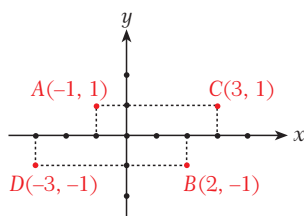
$$9. \vec{PR} = 2 \cdot \vec{AB} \quad 11. \vec{QM} = \vec{PN} = \frac{1}{2} \cdot \vec{DB}$$

13. use the sum of the vectors 14. $|\vec{AB} + \vec{AC}| = 9$ 16. use non-parallel vectors and the vectors $\vec{BA} \parallel \vec{BC}$

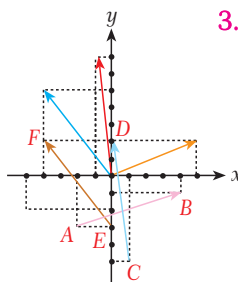
17. use the definition of non-parallel vectors 18. use non-parallel vectors

EXERCISES 2

1.

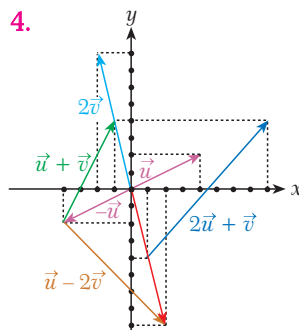


2.



$$3. \sqrt{34}$$

4.



$$5. \text{ a. } \vec{PQ} = (4, -1)$$

$$\text{ b. } \vec{PQ} = (-5, 1)$$

$$\text{ c. } \vec{PQ} = (5, 7)$$

$$\text{ d. } \vec{PQ} = (2, -3)$$

$$6. \vec{a} = (2, 1) \quad \vec{b} = (0, 0)$$

$$7. \text{ a. } \vec{u} + \vec{v} = (3, 9), \vec{u} - \vec{v} = (1, 3), 2\vec{u} = (4, 12), 2\vec{u} - \vec{v} = (3, 9), \vec{u} + 3\vec{v} = (5, 15), -4\vec{u} + 3\vec{v} = (-5, -15)$$

$$\text{ b. } \vec{u} + \vec{v} = (6, 1), \vec{u} - \vec{v} = (-10, 5), 2\vec{u} = (-4, 6), 2\vec{u} - \vec{v} = (-12, 8), \vec{u} + 3\vec{v} = (22, -3), -4\vec{u} + 3\vec{v} = (32, -18)$$

$$\text{ c. } \vec{u} + \vec{v} = (1, -2), \vec{u} - \vec{v} = (1, 2), 2\vec{u} = (2, 0), 2\vec{u} - \vec{v} = (2, 2), \vec{u} + 3\vec{v} = (1, -6), -4\vec{u} + 3\vec{v} = (-4, -6)$$

$$\text{ d. } \vec{u} + \vec{v} = \vec{i} + \vec{j}, \vec{u} - \vec{v} = \vec{i} - \vec{j}, 2\vec{u} = 2\vec{i}, 2\vec{u} - \vec{v} = 2\vec{i} - \vec{j}, \vec{u} + 3\vec{v} = \vec{i} + 3\vec{j}, -4\vec{u} + 3\vec{v} = -4\vec{i} + 3\vec{j}$$

$$\text{ e. } \vec{u} + \vec{v} = -2\vec{i}, \vec{u} - \vec{v} = -4\vec{i} + 2\vec{j}, 2\vec{u} = -6\vec{i} + 2\vec{j}, 2\vec{u} - \vec{v} = -7\vec{i} + 3\vec{j}, \vec{u} + 3\vec{v} = -2\vec{j}, -4\vec{u} + 3\vec{v} = 15\vec{i} - 7\vec{j}$$

$$\text{ f. } \vec{u} + \vec{v} = 6\vec{i} + 6\vec{j}, \vec{u} - \vec{v} = 8\vec{i} + 4\vec{j}, 2\vec{u} = 14\vec{i} + 10\vec{j}, 2\vec{u} - \vec{v} = 15\vec{i} + 9\vec{j}, \vec{u} + 3\vec{v} = 4\vec{i} + 8\vec{j}, -4\vec{u} + 3\vec{v} = -31\vec{i} - 17\vec{j}$$

$$\begin{array}{llll}
 \text{8. a. } |\vec{u}| = \sqrt{10} & \text{b. } |\vec{u}| = \sqrt{5} & \text{c. } |\vec{u}| = \sqrt{13} & \text{d. } |\vec{u}| = 5 \\
 |\vec{v}| = \sqrt{5} & |\vec{v}| = \sqrt{2} & |\vec{v}| = 1 & |\vec{v}| = \sqrt{29} \\
 |2\vec{v}| = 2\sqrt{5} & |2\vec{v}| = 2\sqrt{2} & |2\vec{v}| = 2 & |2\vec{v}| = \sqrt{116} \\
 |\frac{1}{3}\vec{v}| = \frac{\sqrt{5}}{3} & |\frac{1}{3}\vec{v}| = \frac{\sqrt{2}}{3} & |\frac{1}{3}\vec{v}| = \frac{1}{3} & |\frac{1}{3}\vec{v}| = \frac{\sqrt{29}}{3} \\
 |\vec{u} + \vec{v}| = \sqrt{13} & |\vec{u} + \vec{v}| = \sqrt{5} & |\vec{u} + \vec{v}| = 2\sqrt{5} & |\vec{u} + \vec{v}| = \sqrt{106} \\
 |\vec{u} - \vec{v}| = \sqrt{17} & |\vec{u} - \vec{v}| = 3 & |\vec{u} - \vec{v}| = 2\sqrt{2} & |\vec{u} - \vec{v}| = \sqrt{2}
 \end{array}$$

$$\text{9. a. } \vec{v} = 10\sqrt{3}\vec{i} + 10\vec{j} \quad \text{b. } \vec{v} = -15\vec{i} + 15\sqrt{3}\vec{j} \quad \text{c. } \vec{v} = -\frac{\sqrt{2}}{2}\vec{i} - \frac{\sqrt{2}}{2}\vec{j} \quad \text{d. } \vec{u} = -40\sqrt{2}\vec{i} + 40\sqrt{2}\vec{j}$$

$$\text{e. } \vec{v} = 4\cos 10^\circ \cdot \vec{i} + 4\sin 10^\circ \cdot \vec{j} \quad \text{f. } \vec{u} = \frac{\sqrt{3}}{2}\vec{i} - \frac{3}{2}\vec{j} \quad \text{10. } A(10, 9) \quad \text{11. } A(-9, 2) \quad \text{12. } |\vec{u}| > |\vec{v}|$$

$$\text{13. } \frac{\sqrt{17}}{2} \quad \text{14. } 3\vec{i} \text{ km/h} \quad \text{15. } -3\vec{i} + 6\vec{j} \quad \text{16. } -2.4\vec{i} \text{ m/s} \quad \text{17. } 0.4\vec{i} \text{ m/s} \quad \text{18. a. } \vec{v} = -425\vec{i} + 40\vec{j} \quad \text{b. } |\vec{v}| = \sqrt{182225}$$

$$\text{19. speed of the current} = 16\cos 36^\circ \text{ km/h, actual speed of the boat} = 16\sin 36^\circ \text{ km/h} \quad \text{21. } 2 \quad \text{22. } \vec{BC} = \frac{5}{3} \cdot \vec{DC}$$

$$\text{23. } F(2, \frac{5}{2}) \quad \text{24. } \vec{u} = (-10, -2) \quad \text{25. use the definition of linearly dependent vectors} \quad \text{26. } \vec{u} = (\sqrt{6} - \sqrt{2})\vec{v} + (\sqrt{3} - 2)\vec{w}$$

$$\text{28. } \vec{a} + \vec{b} + \vec{c} = -3\vec{a} + 2\vec{b}$$

EXERCISES 3

$$\text{1. a. } 0 \quad \text{b. } -45 \quad \text{c. } 3\sqrt{3} - 1 \quad \text{d. } 2 \quad \text{e. } -304 \quad \text{2. a. } 20 \quad \text{b. } 1 \quad \text{c. } -20 \quad \text{d. } -2 \quad \text{e. } 45 \quad \text{f. } -29 \quad \text{3. } -\frac{3}{2}$$

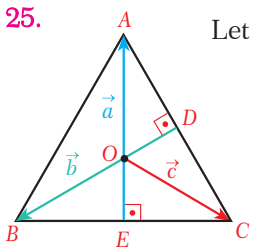
$$\text{4. a. } 6 \quad \text{b. } 37 \quad \text{c. } 13 \quad \text{d. } 132 \quad \text{5. } \sqrt{244} \quad \text{6. a. yes} \quad \text{b. yes} \quad \text{c. yes} \quad \text{d. yes} \quad \text{e. no} \quad \text{f. no}$$

$$\text{7. } m\angle A = 45^\circ, m\angle B = 90^\circ, m\angle C = 45^\circ \quad \text{8. a. } (1, 3) \text{ and } (-1, -3) \quad \text{b. } (3, -4) \text{ and } (-3, 4) \quad \text{c. } (2, -1) \text{ and } (-2, 1)$$

$$\text{9. } t = -\frac{6}{5} \quad \text{10. } \pm\frac{5}{3} \quad \text{11. } -2\pm\sqrt{3} \quad \text{12. yes} \quad \text{13. true, use } \vec{u}^2 = |\vec{u}|^2 \quad \text{14. } -\frac{27}{2} \quad \text{15. } -9 \quad \text{16. use } \vec{u}^2 = |\vec{u}|^2$$

$$\text{17. } (-1, 0) \text{ and } (0, 1) \quad \text{18. use definition of the dot product and } \vec{u} = (u_1, u_2), \vec{v} = (v_1, v_2) \quad \text{19. } 3x + 4y + 7 = 0$$

$$\text{21. } 2 \quad \text{22. } 30 \quad \text{23. } 18 \quad \text{24. find the dot product}$$

25.  Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, $\vec{OC} = \vec{c}$, and $\vec{a} \perp \vec{BC}$, $\vec{b} \perp \vec{AC}$, and use the dot product.

$$\text{26. } d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} \quad \text{27. } 10 \quad \text{28. a. } -\frac{12}{5} \quad \text{b. } -24$$

$$\text{29. a. } 17 \quad \text{b. } 750 \quad \text{30. } 14 \text{ N} \quad \text{31. a. } 10 \text{ N} \quad \text{b. } 10\sqrt{3} \text{ N}$$

$$\text{32. a. } \vec{F} = 22810\vec{i} - 18330\vec{j}$$

$$\text{b. } |\vec{F}| = \sqrt{22810^2 + 18330^2} = \sqrt{856285000}$$



ANSWERS TO TESTS

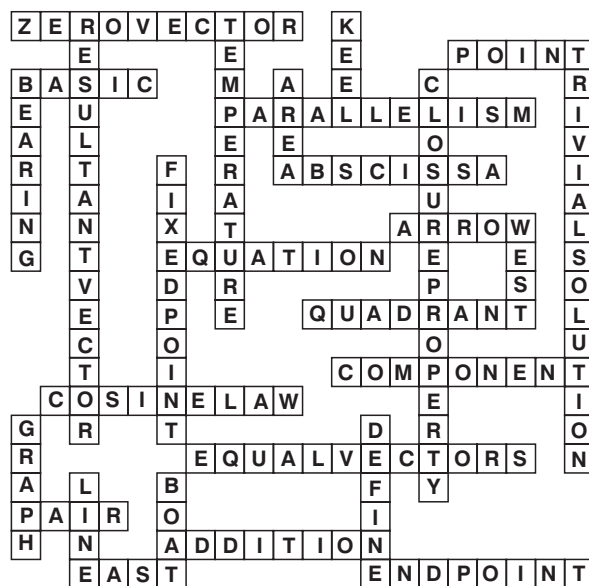
TEST 1

- | | |
|-------|-------|
| 1. A | 11. A |
| 2. A | 12. E |
| 3. D | 13. B |
| 4. A | 14. E |
| 5. C | 15. A |
| 6. E | 16. D |
| 7. C | 17. D |
| 8. B | 18. B |
| 9. C | 19. B |
| 10. C | 20. C |

TEST 2

- | | |
|-------|-------|
| 1. A | 11. A |
| 2. B | 12. D |
| 3. B | 13. B |
| 4. A | 14. C |
| 5. D | 15. D |
| 6. B | 16. B |
| 7. E | 17. A |
| 8. C | 18. D |
| 9. C | 19. E |
| 10. B | 20. C |

SOLUTION TO PUZZLE



GLOSSARY

A

abscissa: the coordinate representing the position of a point along a line perpendicular to the y -axis in a plane.

additive inverse: the negative of a designated quantity.

altitude: the perpendicular distance from the base of a geometric figure to the opposite vertex, parallel side, or parallel surface.

angle: the union of two noncollinear rays with a common endpoint.

area: the size of a flat surface.

arrow: a thin straight stick with a point: $\rightarrow$.

arrowhead: the pointed tip of an arrow.

associative property: an operation is associative if we can apply it to a group of numbers in any order and achieve the same result. For example, $a + (b + c) = (a + b) + c$.

axiom: a basic assumption in mathematics.

B

bearing: an acute angle measured from due north or due south, used to describe the direction of moving ships or vehicles.

C

centroid: the intersection point of the medians in a triangle.

closure property: the property that states: if $a, b \in A$ then $a \odot b \in A$ for an operation $\odot$.

component vector: one of the two horizontal and vertical vectors which have a sum equal to a given vector.

cosine law: in a triangle with side lengths a, b, c , such that A is the angle measure between b and c ,
 $\cos A = (b^2 + c^2 - a^2)/2bc$.

D

diagonal: a segment which joins two nonadjacent vertices of a polygon.

diagram: a graphic representation of an algebraic or geometric relationship.

directed line segment: a line segment with direction.

displacement: the magnitude of a vector from an initial position to a final position.

dot product: the result of adding the numerical product of the corresponding scalar components of two vectors.

E

east: the direction 90° clockwise from north, opposite west.

endpoint: either of two points marking the end of a line segment.

equal vectors: vectors which have the same direction and length.

F

figure: a geometric form consisting of any combination of points, lines, or planes.

fixed point: a point whose position is constant.

force: a vector quantity that tends to produce an acceleration of a body in the direction of its application.

H

horizontal: parallel to the horizon, or in the plane of the horizon.

I

identity element: the element of a set that when combined with another element in a particular operation leaves that element unchanged. For example, the zero vector $\vec{0}$ is the identity element for vector addition.

inclined: sloping, slanting, or leaning.

initial point: the starting point of a vector.

intersect: to cut across or through.

intersection: the act, process, or result of intersecting.

L

linear combination of vectors: a vector expressed in terms of other vectors.

N

natural number: any number which is a member of the set $\{1, 2, 3, \dots\}$.

navigation: the act or practice of sailing a ship.

negative: less than zero.

norm of a vector: another name for the length of a vector.

north: the direction along a meridian 90° counterclockwise from east; the direction to the left of sunrise.

notation: any series of signs or symbols used to represent quantities or elements in mathematics.

numerator: the dividend of a fraction.

O

opposite vectors: two vectors which have opposite direction and the same length.

ordinate: in the rectangular coordinate system, the coordinate representing the distance from a specified point to the x -axis, measured parallel to the y -axis.

origin: the point of intersection of the coordinate axes in the rectangular coordinate system.

P

parallelism: being parallel.

parallelogram: a four-sided plane figure with two opposite parallel sides .

parallelogram method: to add vectors using the parallelogram method, we make a parallelogram by drawing two additional sides, each passing through the terminal point of one of the vectors and parallel to the other vector. The sum is the diagonal from the common initial point to the intersection of the two lines.

perpendicular: two vectors are perpendicular if they form a right angle.

perpendicularity: the state of being perpendicular.

plot: to locate (points or other figures) on a graph by means of coordinates.

polygon: a closed plane figure bounded by three or more line segments.

polygon method: to add vectors using the polygon method, we join the initial point and terminal point of each successive vector. The sum is the vector whose initial point is the initial point of the first vector and whose terminal point is the terminal point of the last vector.

position vector: a vector which represents another vector, and has its initial point at the origin.

Pythagorean theorem: a theorem stating that the sum of the squares of the lengths of the sides of a right-angle triangle is equal to the square of the length of the hypotenuse.

Q

quadrant: any of the four areas into which a plane is divided by the reference axes in the rectangular coordinate system, designated first, second, third, and fourth, counting counterclockwise from the area in which both coordinates are positive.

quadrilateral: a polygon with four sides.

R

rectangle: a four-sided plane figure with four right angles.

resultant vector: a single vector that is the equivalent of a set of vectors.

rhombus: a quadrilateral whose sides are all equal length.

S

scalar: a quantity which has only magnitude, for example, length or speed.

scalar component: the numbers v_1 and v_2 for the vector $\vec{v}$ in the directions of $\vec{i}$ and $\vec{j}$.

shore: the low land along the edge of an ocean, sea, lake, or river.

side: a line or segment which bounds a plane figure.

sketch: a simple, quickly-made drawing.

south: the direction 90° clockwise from east and directly opposite north.

standard base vectors: horizontal and vertical unit vectors which start at the origin.

subtraction of vectors: the operation of adding a vector to the opposite of another vector.

T

terminal point: the finishing point of a vector.

theorem: a proposition that can be proved on the basis of explicit assumptions.

triangle: a three-sided polygon.

triangle inequality: The property that states:

$$|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|.$$

trivial solution: the solution of an equation in which every variable is equal to zero.

U

unit vector: a vector with length one unit.

V

variable: a symbol representing a quantity. For example, in the expression $a^2 + b^2 = c^2$, a , b , and c are variables.

vector: a quantity which has both magnitude and direction, for example, velocity, or movement along a bearing.

velocity: a vector quantity whose magnitude is a body's speed and whose direction is the body's direction of motion.

vertical: at right angles to the horizon; upright.

W

west: the direction 270° clockwise from north and directly opposite east.

Z

zero vector: a vector with length zero, written $\vec{0}$.

